

Technical Activity Report

A Multi-Phase Investigation of Visible Corrosion and Structural Condition in
Ferrocement Specimens

Draft working document — not for distribution

Draft in progress

Abstract

Corrosion of steel reinforcement is a principal durability concern for cementitious structural elements, and repeated photography offers an inexpensive, non-destructive way to observe it over time. This activity uses a predecessor experimental dataset of 48 ferrocement specimens tested across two exposure campaigns and photographed repeatedly; in the current audited basis, 791 aligned, readable observations carry dense visible-corrosion labels, while wire-area loss and ultimate load are each obtained only once per specimen, at the end of its exposure period. Built around this asymmetry between dense surface supervision and sparse terminal structural supervision, the activity progressed through classical, deep-embedding, and interpretable image representations of visible corrosion; a feasibility study of hidden structural-condition inference; robustness evaluation under grouped and transfer-based holdout regimes; a refocused study of terminal ultimate load; and an exploratory degradation and proxy-remaining-useful-life screening chain. Visible surface corrosion proved the strongest and most consistently learnable image-based quantity, although specific benchmarks carry a label-adjacency risk and treatment-level transfer robustness proved phase-dependent. Hidden structural inference was substantially weaker and less transferable, and must be interpreted in light of its sparse, terminal, campaign-confounded supervision. Image-derived features did not provide reliable incremental predictive value over metadata for terminal ultimate load under pooled evaluation. Moving to treatment- and campaign-level transfer materially changed which conclusions the evidence could support, with campaign holdout acting as a severe extrapolation stress test given how closely campaign identity is entangled with specimen design here. The degradation and proxy-remaining-useful-life work is methodological and diagnostic rather than validated prognosis: no true remaining-useful-life supervision exists in the dataset. A leakage-safe dataset for a subsequent four-class severity classification task was also prepared, though not yet modelled. The activity’s principal contribution is a rigorously bounded account of what repeated corrosion imagery can and cannot support here, together with a traceable basis for the work that should follow.

Executive Summary

Purpose and evidence basis. This report documents an MSc technical activity investigating how much can be learned about a ferrocement specimen’s condition from repeated corrosion photography. It builds on a predecessor experimental dataset — 48 specimens fabricated, chloride-exposed, and terminally tested across two campaigns, external to the present activity — and works from the resulting photographic record, specimen metadata, and terminal mechanical measurements. Visible-corrosion labels are available for nearly every image in the current audited basis (791 of 792 raw files), while wire-area loss and ultimate load are each obtained only once per specimen, at the end of its exposure period. This asymmetry between dense surface supervision and sparse terminal structural supervision governs how every finding in this report should be read.

Technical activity. The activity comprised: an audit and alignment of the raw photographic and workbook data; three successive approaches to visible-corrosion estimation (a classical hand-crafted baseline, a frozen deep-image embedding, and an interpretable multi-family feature representation); a feasibility study of hidden structural-condition inference; systematic robustness evaluation under grouped, treatment-held-out, and campaign-held-out specimen partitions; a refocused study explicitly testing whether images add value beyond metadata for terminal ultimate load; an exploratory degradation-curve and proxy-remaining-useful-life screening pipeline; preparation of a leakage-safe dataset and augmentation pipeline for a four-class corrosion-severity classification task; and an audit of the project’s own software and reproducibility practices.

Main findings. The evidence supports a clear hierarchy of confidence:

1. **Visible corrosion** is the strongest and most consistently learnable image-based quantity in this dataset, across three distinct representations.
2. **Hidden structural inference** is substantially weaker: the available structural models are either mixed-input or, in their final, most carefully selected form, metadata-only, and none transfers reliably across campaigns.
3. **Terminal ultimate load**, one of the two directly measured terminal structural endpoints and the endpoint selected for the refocused analysis, is dominated by specimen and exposure metadata under pooled evaluation; no robust incremental benefit from image-derived features was established.
4. **Degradation and proxy-remaining-useful-life screening** remain exploratory: the pipeline is a working demonstration of a derivation chain, not a validated prognostic method.

Two cross-cutting points qualify all four: which evaluation regime is used materially changes what can be claimed, and leave-one-campaign-out is a severe extrapolation stress test in this dataset because campaign identity is entangled with steel-mesh configuration, chloride level, and exposure duration. In addition, some strong visible-corrosion benchmarks rely on image-derived

features that are very closely aligned with the target, creating shortcut or label-adjacency risk; the strongest case is the near-reconstructive total-rust pairing identified by the later feature audit, and this is checked and flagged explicitly where it applies.

Scope and limitations. Every structural or prognostic conclusion in this report is built on 48 terminal structural measurements, from only two experimental campaigns whose design factors are almost perfectly confounded with campaign identity. No specimen has a repeated or intermediate structural measurement, and no specimen has an observed failure or lifetime event; structural values reported at non-terminal weeks are always model estimates, never measurements. The four-class classification branch’s data preparation is complete, but no classifier has been trained, compared, or evaluated at the time of writing.

Current status and next direction. The main report body (dataset audit, methodological framework, activity chronology, integrated results, interpretation, software/reproducibility audit, limitations, and conclusions) is complete. Reproducibility has improved over the course of the project — deterministic augmentation, saved split manifests, and an automated feature-diagnostics report are already in place — though two live configuration and source-code issues, identified and located in this report, still prevent an unmodified rerun of some pipeline stages. The most immediately justified next modelling step is training a baseline classifier for the prepared four-class branch, evaluated with specimen-disjoint, class-sensitive metrics. Beyond that, materially stronger structural or remaining-life claims will ultimately require new experimental evidence — additional independent campaigns, decoupled design factors, more structural-labelled specimens, repeated structural measurements, and observed lifetime data — rather than further model refinement alone.

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Chapter 1

Introduction and Project Scope

1.1 Background and Motivation

Corrosion of steel reinforcement is a principal durability concern for reinforced cementitious structural elements, and ferrocement — a thin, densely mesh-reinforced cementitious composite — is no exception. Assessing that corrosion typically requires either destructive sampling or specialised instrumentation, both of which are costly to apply repeatedly over an element’s service life. Repeated photography of an exposed surface is, by contrast, comparatively inexpensive and non-destructive, and it raises a natural question: how much can be inferred about a specimen’s condition from its accumulated photographic record alone?

Answering that question requires distinguishing two quantities that are easy to conflate but evidentially very different. Visible surface corrosion is directly observable in a photograph and can, in principle, be labelled and re-labelled from the same image at any time. Internal structural condition, by contrast, is not observed longitudinally in the photographic record at all: structural supervision is available only through terminal measurements — wire-area loss and ultimate load — obtained once per specimen, destructively, at the end of its exposure period. This report’s central technical thread, developed across the chapters that follow, is this distinction: what a dense, repeatedly-observed surface quantity can support scientifically is not automatically what sparse, terminal structural measurements can support, and the two must be evaluated, and reported, differently.

The physical specimens, their fabrication, their accelerated chloride-exposure programme, and their terminal mechanical testing are predecessor experimental work, external to the present activity, documented in an earlier MSc thesis [Hossain, 2025] and a related conference paper [Sorrentino et al., 2025]. The present activity builds on, and uses, the experimental dataset generated by that preceding work; it does not repeat the physical experiment. Some of the later image-feature work in this activity is best described as a Python reconstruction and extension of concepts from that predecessor workflow, rather than an exact reproduction of it (Chapter 4). Facts about the physical experiment are attributed to that predecessor source throughout this report; facts about how the resulting data has been organised, audited, modelled, and interpreted are attributed to the present activity.

1.2 Project Context and Data Basis

The dataset comprises 48 ferrocement specimens tested across two experimental campaigns, each photographed repeatedly over its exposure period. Visible-corrosion labels are available throughout this photographic record — in the current audited basis, 791 aligned, readable observations out of 792 raw image files (Chapter 2) — while terminal structural measurements exist for each specimen exactly once, at the end of its exposure period. This asymmetry between dense, repeatedly observed visible-corrosion supervision and sparse, single-timepoint structural

supervision is the single most consequential structural fact about this dataset, and it is why visible-corrosion estimation and structural-condition inference are treated as scientifically distinct problems throughout this report rather than as two readings of the same underlying task. Chapter 2 presents the full dataset audit; it is not repeated here.

1.3 Project Objectives and Research Questions

The activity was organised around five research questions, each answered, to varying degrees of confidence, by the work reported in Chapters 4 and 5.

RQ1. To what extent can visible surface corrosion be estimated from repeated images, using classical, deep-embedding, and interpretable representations in turn?

RQ2. Can hidden structural condition, or terminal structural response specifically, be inferred reliably from the available image-derived and metadata inputs?

RQ3. Do image-derived features provide incremental predictive value beyond known specimen and exposure metadata for terminal ultimate load?

RQ4. How robust are the resulting conclusions when evaluation moves from a grouped specimen holdout to treatment-level and campaign-level transfer tests?

RQ5. What can degradation-curve fitting and threshold-based proxy-RUL screening legitimately contribute, given the absence of repeated structural measurements and of any observed true lifetime outcome?

A later, separate workstream — preparing a leakage-safe dataset and augmentation pipeline for four-class visible-corrosion severity classification — is deliberately not framed as a sixth research question here, since it addresses data preparation rather than a modelling question, and no classifier has yet been trained on it (Section 1.4).

1.4 Scope of the Technical Activity

The activity reported here comprises: an audit and alignment of the raw photographic and workbook data; a classical visible-corrosion baseline; a frozen deep-image-embedding experiment; an engineered, interpretable image feature representation; structural-condition feasibility modelling; robustness evaluation under grouped, leave-one-treatment-out, and leave-one-campaign-out regimes; degradation-curve fitting and proxy-RUL screening; a refocused study of terminal ultimate load; preparation of an augmented, specimen-grouped dataset for four-class classification; and an audit of the project’s own software and reproducibility practices. Chapter 4 describes each of these in turn; Chapter 7 covers the last.

Several boundaries on what this report claims are stated here explicitly, because they govern how every later chapter should be read. This report does not establish, and does not claim, a validated remaining-useful-life model: no true RUL supervision exists in the dataset, and the proxy-RUL construct introduced in Chapter 3 is an explicitly qualified screening quantity, not a lifetime prediction. It does not claim that visible surface corrosion causally determines structural capacity; where a statistical association between the two is reported, it is reported together with the confounding structure that limits how it may be interpreted. It does not produce an externally validated or deployment-ready model of any kind. It does not include a trained four-class classifier: that branch’s data preparation is complete, but no classifier has been trained, compared, or evaluated (Chapter 4). And it does not treat model-estimated structural states at non-terminal observation weeks as measurements — they are, throughout, the output of a model, not a repeated physical observation.

1.5 Main Contributions of the Activity

The activity’s principal, completed contributions are as follows.

(A) Audited and aligned a specimen-level analysis basis from the raw photographic and workbook data, resolving several version-specific discrepancies across the project’s successive implementation phases (Chapter 2).

(B) Established visible-corrosion estimation as the most consistently learnable image-based task in this dataset, while identifying specific label-adjacency and near-tautology risks that limit how some individual benchmarks should be read (Chapter 4, Chapter 5).

(C) Quantified the substantially weaker and less stable transferability of hidden structural prediction, and disentangled the contribution of image-derived inputs from metadata/campaign-linked predictive structure through explicit ablations and transfer tests (Chapter 4, Chapter 5, Chapter 6).

(D) Refocused the structural research question onto terminal ultimate load — one of the two directly measured terminal structural endpoints, and the one selected for this analysis — and explicitly tested, rather than assumed, whether images add predictive value beyond metadata for it (Chapter 4).

(E) Built and critically evaluated a degradation-curve and proxy-RUL screening chain across two independently configured generations, establishing why its contribution is methodological and diagnostic rather than validated prognosis (Chapter 4, Chapter 6).

(F) Prepared a leakage-safe, specimen-grouped, reproducible dataset and augmentation pipeline for a four-class visible-corrosion severity classification task, ready for subsequent model training (Chapter 4).

(G) Documented concrete reproducibility and software-engineering issues in the project’s own codebase, and maintained a traceable, source-cited evidence record for every quantitative claim in this report (Chapter 7).

None of these is presented as a novel publication-level scientific contribution in its own right; each is a specific, bounded piece of work completed during this activity, reported at the level of confidence its own evidence supports.

Read together, the activity’s findings are, at a high level: visible surface corrosion is the most consistently learnable image-based quantity available in this dataset; hidden structural inference is substantially weaker, and strongly affected by sparse supervision and campaign-linked design structure; images do not show a reliable pooled incremental benefit over metadata for terminal ultimate load under the settings tested here; and the proxy-RUL construct remains a useful exploratory screening tool rather than a validated prognosis of remaining service life. The evidence and qualifications behind each of these statements are given in full in Chapters 4–6.

1.6 Report Organisation

Chapter 2 audits the dataset and establishes the supervision asymmetry this introduction has previewed. Chapter 3 sets out the methodological framework — evaluation regimes, representation strategies, and the observed/model-derived/derived-screening distinction — shared across later chapters. Chapter 4 reconstructs the project’s successive implementation phases chronologically. Chapter 5 integrates that evidence by scientific question rather than by phase, and Chapter 6 explains what the integrated evidence means and why it justified the research decisions taken. Chapter 7 documents the project’s software and reproducibility practices and their current limitations, Chapter 8 states the limitations of the evidence itself by category, and Chapter 9 gives the current project status and the work this report recommends next.

Chapter 2

Dataset, Experimental Context, and Data Audit

This chapter sets out the physical experimental context, the structure of the available data, and the outcome of a dataset-level audit that later chapters build upon rather than re-deriving. It distinguishes between facts about the physical experiment, which originate from predecessor work external to the present project, and facts about how that experimental record has been organised, cleaned, and represented within the present project’s own pipeline.

2.1 Experimental Context

The dataset originates from a physical durability study of ferrocement specimens subjected to accelerated chloride-induced corrosion. Ferrocement elements reinforced with steel mesh were fabricated, exposed to a saline environment for an extended period, and photographed repeatedly to record the progression of visible surface corrosion. At the end of each specimen’s exposure period, a terminal mechanical test was carried out to characterise its residual structural condition. The physical fabrication, exposure, and testing programme was designed and executed as predecessor work, documented in an earlier MSc thesis [Hossain, 2025] and a related conference paper [Sorrentino et al., 2025]; the present project did not conduct this physical experiment. It instead works from the photographic record, the associated specimen/exposure metadata, and the terminal mechanical-test measurements that resulted from it, and reconstructs a specimen-to-campaign/treatment mapping from specimen identifiers, cross-checked against the treatment tables reported in the predecessor thesis.¹ Facts concerning the physical experiment itself are attributed to this predecessor source; facts concerning how the present project has organised or audited the resulting data are attributed to the present project’s own code, configuration, or generated outputs.

The experimental programme comprises two campaigns, executed at different times and, notably, at different steel-mesh and chloride-exposure settings. The first campaign involved 24 specimens organised into five treatment series, reinforced with a higher steel-mesh configuration and exposed to a lower chloride concentration; specimens from this campaign were terminally tested after 196 days of ageing (week 28 of observation). The second campaign involved a further 24 specimens organised into four treatment series, reinforced with a lower steel-mesh configuration and exposed to a higher chloride concentration; specimens from this campaign were terminally tested after 252 days of ageing (week 36 of observation). Table 2.1 summarises the resulting design matrix, and Section 2.6 discusses a structural consequence of this design that later chapters must account for when interpreting results.

¹The specimen-to-campaign and specimen-to-treatment mapping used throughout the present project’s pipeline is maintained as a single, version-controlled table that is documented as the preferred source for this reconstruction, in preference to the raw treatment columns present in the data workbook.

Across both campaigns, seven coarse treatment protocols are represented: an untreated control; a corrosion inhibitor mixed into the mortar matrix at casting; a corrosion inhibitor applied to the surface after casting (applied by different methods — general surface application in the first campaign, spray or brush application in different series of the second campaign); a protective surface paint; a combined paint-plus-surface-inhibitor treatment; and two treatments incorporating a glass-based surface compound, applied either alone or together with the surface inhibitor. These seven protocols are unevenly represented, both overall and within each campaign, as shown in Table 2.1.

Terminal structural condition was assessed through a single mechanical test per specimen. Two structural quantities are reported in the predecessor documentation: the peak load recorded during a four-point bending test to the specimen’s load-bearing limit, and the loss of cross-sectional area of the outermost longitudinal reinforcing wires near the aged surface, obtained from microscopic examination of a cross-section taken at the failure location [Hossain, 2025].² Both quantities are recorded once per specimen, at the single terminal time point associated with that specimen’s campaign.

Table 2.1: Experimental campaign and design matrix. Steel-mesh configuration, chloride concentration, and terminal testing time are recorded once per specimen and do not vary within a campaign. Treatment-group specimen counts are reconstructed from the project’s specimen-mapping table.

Campaign	Specimens	Steel-mesh configuration	NaCl concentration	Terminal week (observation)	Terminal ageing (days)
Campaign 1	24	7	3.5%	28	196
Campaign 2	24	4	5.0%	36	252

Campaign	Specimen count by treatment protocol							Total
	Control (NO)	Mixed-in inhibitor (MI)	Surface inhibitor (SA)	Paint (PA)	Paint+ surface (SA_PA)	Surface+ glass (SA_VF)	Glass compound (VF)	
Campaign 1	4	6	6	1	3	2	2	24
Campaign 2	6	–	12	6	–	–	–	24
Total	10	6	18	7	3	2	2	48

2.2 Data Sources and Modalities

The material available to the present project is organised into four distinct modalities, which differ substantially in their granularity and availability and must not be conflated.

A. Image observations. Photographs of each specimen’s surface, taken repeatedly over its observation period. Each image is associated with a specimen identifier, an acquisition date, and an observation week, encoded in the image filename. This is the only modality that is genuinely longitudinal at fine time resolution (Section 2.3).

B. Specimen and exposure metadata. Design and exposure variables recorded once per specimen and constant across that specimen’s entire observation period: campaign identity, treatment series and protocol, steel-mesh configuration, and chloride concentration (Table 2.1). A small number of additional fields, such as observation week itself, vary by image rather than by specimen.

²This attribution follows the present project’s own audit documentation, which cites the relevant thesis chapters for these two definitions; the thesis itself was not re-read in full for the present chapter.

C. Visible-corrosion labels. Manually assigned quantitative and categorical measures of surface corrosion, recorded for essentially every usable image: a continuous total (whole-specimen) rust percentage, a continuous peak (localised maximum) rust percentage, and, in the currently audited data table, a four-class ordinal severity category for each of these two measures. These labels are therefore available at the same fine time resolution as the images themselves. As discussed in Section 2.5, the number of ordinal levels used for these severity categories is itself not constant across the project’s successive dataset builds.

D. Structural endpoint measurements. The two terminal mechanical quantities introduced in Section 2.1 — wire-area loss and ultimate load — recorded once per specimen, at that specimen’s single terminal observation.

The essential asymmetry that this chapter establishes, and that Section 2.4 develops in detail, is that modalities A and C are dense (available for nearly every image observation), while modality D is sparse (available for only one observation per specimen). Modality B sits between the two: it is constant per specimen but known for every observation of that specimen. Table 2.2 summarises the granularity and availability of each modality, including the class distributions of the two four-class visible-corrosion severity categories, which are markedly imbalanced toward the lowest-severity class.

Table 2.2: Data modalities and label availability, computed directly from the aligned 791-row dataset table.

Modality	Content	Granularity	Availability
A. Image observations	Specimen surface photograph	Per image	791 of 792 aligned observations
B. Specimen/exposure metadata	Campaign, treatment, steel-mesh configuration, chloride concentration, observation week	Per specimen (design variables); per image (week)	All 791 aligned observations
C. Visible-corrosion labels	Total and peak rust percentage (continuous); four-class severity category for each	Per image	791 of 792 aligned observations. Total-category counts: 658 / 99 / 20 / 14 (classes 1–4); peak-category counts: 526 / 107 / 95 / 63
D. Structural endpoint measurements	Wire-area loss; ultimate load	Per specimen (single terminal observation)	48 of 791 aligned observations (one per specimen)

2.3 Longitudinal Observation Structure

Each specimen was photographed repeatedly over its observation period, yielding, after the alignment audit described in Section 2.5, between 15 and 18 image observations per specimen (median 16). Observation weeks are not sampled on a strict fixed weekly cadence: across the aligned dataset, images were captured at twenty-five distinct week values spanning week 0 to week 36, with irregular spacing between successive observations of the same specimen.

The repeated images belonging to a given specimen are not independent observations in

the sense required by a naive statistical treatment. They share a single physical specimen, a single fabrication and treatment history, and a single acquisition and imaging context, and they trace out one continuous corrosion trajectory rather than twenty-five unrelated draws from a population of unrelated images. An observation at week 12 of a given specimen is informative about that same specimen’s observation at week 15 in a way that it is not informative about a different specimen’s observation at week 15.

This has a direct implication for how the data may legitimately be used to evaluate a predictive model, which the present chapter states only in general terms: any evaluation procedure that treats individual image rows as exchangeable, independent units risks allowing information about a specific specimen to pass from a training partition to a test partition through a different image of the same specimen, rather than through genuine generalisation to unseen specimens. The specific evaluation procedures the project adopted in response to this concern are the subject of Chapter 3.

2.4 Structural Supervision

The dataset comprises 48 specimens in total. Each specimen contributes exactly one structural observation — its terminal wire-area-loss and ultimate-load measurement, taken at that specimen’s single terminal test (week 28 for Campaign 1 specimens, week 36 for Campaign 2 specimens; see Table 2.1). No specimen contributes more than one structural observation, and no specimen’s structural condition is measured at any intermediate observation week.

This makes structural supervision fundamentally different in kind from the dense visible-corrosion labels described in Section 2.2. Visible-corrosion labels describe a trajectory: for a given specimen, they are available at many points along its observation period, and changes in visible corrosion over time can be observed directly. Structural measurements describe a single endpoint: for a given specimen, nothing is known about its structural condition before the terminal observation, and nothing is known about it after.

A direct consequence is that the dataset does not contain the information required to pose intermediate structural condition, a structural trajectory, or a time-to-failure quantity as a directly supervised learning problem. Doing so would require either repeated structural measurements of the same specimen over time, or a recorded failure event time, and neither is present. Table 2.3 summarises this distinction, separating quantities that are directly observed in the data from quantities that later phases of the project construct through modelling, quantities that are further derived from those modelled quantities for screening purposes, and quantities that are not available at all.

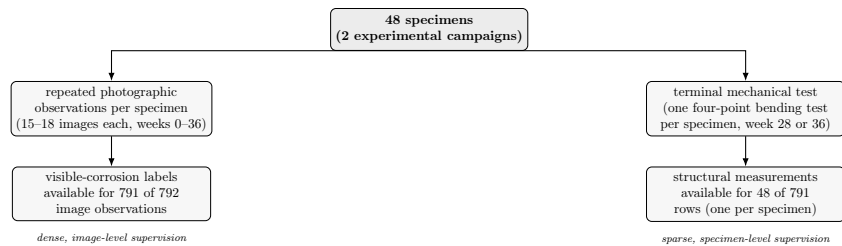


Figure 2.1: Schematic contrast between dense, image-level visible-corrosion supervision and sparse, specimen-level structural supervision, following directly from Table 2.3. This is a report schematic illustrating the data structure described in Sections 2.2–2.4; it is not a plot of experimental results.

Table 2.3: Directly observed, model-derived, derived-screening, and unavailable quantities. This distinction is maintained throughout the report and must not be blurred: a model-derived estimate of intermediate structural condition is not a measurement, and a derived screening output is not a supervised prediction target.

Category	Quantities
Directly observed	Image; observation week; visible-corrosion labels (total and peak rust percentage and their four-class categories); terminal structural measurements (wire-area loss, ultimate load) at each specimen’s single terminal observation.
Model-derived	Estimated hidden structural condition at intermediate (non-terminal) observation weeks; fitted specimen-level degradation trajectories over the observation period.
Derived screening outputs	Threshold-crossing time computed from a fitted degradation trajectory (“proxy remaining-useful-life”); see Section 3.8.
Not available	A directly supervised remaining-useful-life or time-to-failure target; an experimentally observed failure event time. No evidence encountered in the reviewed project documentation or the predecessor thesis material establishes such a target for this dataset.

2.5 Dataset Audit and Alignment

The image archive contains 792 image files in total, spanning 48 specimens. Of these, one file — a single image belonging to a Campaign 2 specimen from the surface-inhibitor (spray) series — is corrupted and cannot be decoded by standard image-reading routines. The remaining 791 files are readable and align to a corresponding metadata record. Table 2.4 summarises the resulting dataset dimensions on the aligned (791-row) basis that this report adopts by default.

Table 2.4: Dataset overview, aligned (791-row) basis, computed directly from the current canonical dataset table.

Quantity	Value
Image files (raw archive)	792
Corrupted / unreadable image files	1
Aligned rows adopted by this report	791
Specimens	48
Experimental campaigns	2
Observation week range	0–36 (irregular; 25 distinct week values)
Image observations per specimen	15–18 (median 16)
Specimens with structural endpoint measurements	48 of 48 (one observation each)

The project’s own row-count convention for this one corrupted file has evolved across its successive implementation phases, and this evolution is best understood as a progressive tightening of data-quality policy rather than as an unresolved inconsistency. Table 2.5 summarises the three distinct policies adopted. The earliest exploratory implementation removed the affected row from its working table without substitution, arriving at 791 usable rows. The next three implementation phases each instead retained all 792 rows and substituted the missing image-derived quantities — handcrafted features in one phase, a learned embedding in another,

a broader interpretable feature set in a third — with the corresponding column-wise median value, so that the corrupted observation remained present but carried an imputed rather than a measured image representation. The current and subsequent implementation phases instead apply an explicit, configuration-level data-quality rule that excludes any image that cannot be aligned to a readable, matched record, which is the policy this report adopts by default: 791 aligned rows drawn from 792 image files, 48 specimens.

Table 2.5: Handling of the single corrupted image observation across successive implementation phases, verified directly against each phase’s own data-loading code.

Phase	Resulting row count	Handling mechanism
Earliest exploratory implementation	791	Row silently omitted during table construction; no substitution.
Classical baseline	792	Row retained; affected hand-crafted image features replaced by column-wise medians.
Deep-embedding phase	792	Row retained; affected learned image embedding replaced by column-wise medians.
Interpretable multi-family feature phase	792	Row retained and flagged; affected feature set (91 features) replaced by column-wise medians.
Current canonical phase (adopted here), inherited by the later classification/augmentation effort	791	Row excluded by an explicit alignment/readability policy prior to feature extraction.

Because this row-count convention differs across phases, any quantitative result drawn from an earlier phase and discussed later in this report (Chapter 4) is understood to be computed on a 792-row basis with one imputed observation, whereas results drawn from the current canonical phase and the later classification effort are computed on a 791-row aligned basis. The two bases are not row-for-row identical and are not treated as interchangeable when results are compared (Chapter 5).

A second, distinct schema change concerns the visible-corrosion severity category labels introduced in Section 2.2. The currently audited data table represents both the total-rust and peak-rust severity categories on a four-level ordinal scale. An earlier canonical dataset build, produced during the interpretable multi-family feature phase on the historical 792-row basis, instead represented the same two quantities on a five-level ordinal scale, with materially different class counts (Table 2.6). The data-loading code from that earlier phase confirms the change directly: it is written to match a raw workbook column defined over a five-level range, where the currently audited workbook instead defines the corresponding column over a four-level range. This is a genuine change in the underlying label schema between the workbook snapshot used at that phase and the workbook snapshot audited here, not a data-quality defect internal to the project, and the two schemas are not treated as interchangeable in this report: a categorical-classification result reported for the earlier phase in Chapter 4 refers to a five-class problem and is not described as an evaluation of the four-class scheme that the current dataset, and the later classification and augmentation effort (Section 3.4, Chapter 4), target.

Table 2.6: Visible-corrosion severity category schema across dataset builds. Class counts are ordered from the lowest to the highest severity level.

Dataset build		Row-count basis	Total-rust category counts	Peak-rust category counts
Interpretable family feature (historical)	multi-phase	792 (imputed)	5 classes: 493 / 51 / 38 / 77 / 133	5 classes: 304 / 159 / 72 / 122 / 135
Current canonical phase (audited; adopted by this report)		791 (aligned)	4 classes: 658 / 99 / 20 / 14	4 classes: 526 / 107 / 95 / 63

2.6 Dataset Structure and Potential Confounding

Table 2.1 shows that steel-mesh configuration, chloride concentration, and terminal ageing duration do not vary independently of campaign identity within the observed sample: every specimen with the higher steel-mesh configuration also has the lower chloride concentration and the shorter terminal ageing duration, and belongs to Campaign 1; every specimen with the lower steel-mesh configuration also has the higher chloride concentration and the longer terminal ageing duration, and belongs to Campaign 2. No specimen in the dataset combines a Campaign-1 mesh configuration with a Campaign-2 chloride level, or vice versa.

This structural property of the experimental design has a direct consequence for how any statistical association between a design-linked quantity and a measured outcome must be interpreted. Where an association is estimated by pooling specimens across both campaigns, that association cannot, on the evidence of the pooled sample alone, be uniquely attributed to any one of the co-varying factors — campaign identity, mesh configuration, chloride exposure, and ageing duration are, within this sample, different labels for largely the same partition of specimens. A pooled correlation involving one of these factors, or a quantity correlated with one of them, should accordingly be read as a description of the pooled sample rather than as evidence isolating the effect of that factor. This report returns to the practical consequences of this property for model evaluation in Chapters 3 and 5; the present section deliberately stops short of discussing any modelling result.

A related consequence concerns what it means to evaluate a model by holding out an entire campaign. Because campaign identity coincides with a specific combination of mesh configuration, chloride level, and ageing duration, holding out one campaign does not test a model against unseen specimens drawn from the same underlying design space as its training data; it tests the model against specimens whose design characteristics differ systematically, and simultaneously along several correlated dimensions, from every specimen the model was trained on. Such an evaluation is better characterised as a test of extrapolation across the experimental design space than as an ordinary test of generalisation to unseen specimens within that space. This distinction is developed formally in Section 3.3.

2.7 Summary of Dataset Constraints

The dataset available to this project is constrained in ways that directly shape the modelling strategies described in later chapters:

- Visible-corrosion supervision is dense (available for 791 of 792 image observations) while structural supervision is sparse (available for exactly 48 of those observations, one per specimen).

- Repeated image observations of the same specimen are correlated, not independent, and must be treated at the specimen level for evaluation purposes.
- No repeated or intermediate structural measurement exists for any specimen; the dataset supports estimation of a single terminal structural state per specimen, not a directly observed structural trajectory or failure time.
- Row-count and missing-observation conventions differ across the project’s implementation phases and must be stated explicitly whenever a quantitative result is cited.
- Campaign identity, steel-mesh configuration, chloride concentration, and terminal ageing duration co-vary near-perfectly across the observed specimens, limiting what can be inferred from pooled associations and giving cross-campaign evaluation the character of extrapolation rather than ordinary generalisation.

These constraints are the principal reason the project’s modelling strategy, described next in Chapter 3, treats specimen-level grouping, multiple evaluation regimes, and a conservative distinction between measured, modelled, and derived-screening quantities as required elements of the methodology rather than optional refinements.

Chapter 3

Common Methodological Framework

Chapter 2 established the structure and constraints of the available data. This chapter describes the methodological framework that the project’s successive implementation phases used, with variation, to work within those constraints. Rather than repeating a phase-by-phase account here, it extracts the building blocks that recur across phases — evaluation regimes, representation strategies, model families, metrics, and the treatment of remaining-useful-life — so that Chapter 4 can describe each phase’s activity without re-explaining shared machinery, and so that later comparisons (Chapter 5) can be made on a consistent methodological footing.

3.1 General Modelling Philosophy

The project’s successive phases addressed several distinct prediction problems over the same underlying dataset: estimating a specimen’s current visible-corrosion state from an image; predicting a specimen’s terminal structural condition; fitting an exploratory model of how corrosion and structural condition evolve over time; screening for an approximate service-life indicator; and, in a later and separate effort, classifying images into a discrete visible-corrosion severity category. These are not variations on a single problem, and the report does not treat them as such.

What is shared across phases is not the prediction target but a methodological principle that later phases articulated increasingly explicitly: a claim made about the data must correspond to the supervision the data actually provides. A model trained on dense visible-corrosion labels supports claims about visible-corrosion estimation; it does not, by itself, support claims about structural condition, service life, or failure risk, however plausible an extrapolation from corrosion to those quantities might seem. The distinctions drawn in Section 2.7 between directly observed, model-derived, and derived-screening quantities are the practical expression of this principle, and Section 3.8 returns to its most consequential application.

3.2 Leakage-Safe Specimen-Level Evaluation

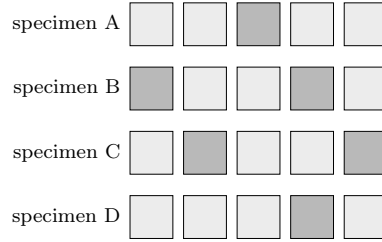
Section 2.3 established that the repeated image observations of a given specimen are correlated rather than independent. This has a direct and easily overlooked consequence for evaluation.

Consider a dataset of image observations split into training and test partitions by assigning individual image rows at random, without regard to which specimen each row belongs to. Because a typical specimen contributes fifteen to eighteen images spanning its entire observation period, a random row-wise split will, with high probability, place most of a given specimen’s images in the training partition and a few of the same specimen’s images in the test partition. A model evaluated in this way can achieve a low test error not by learning corrosion cues that generalise to specimens it has never seen, but by learning to recognise that specific specimen — its particular lighting, its particular background, or idiosyncrasies of its individual corrosion

pattern — from the many other images of it already seen in training. The resulting test error understates the error the same model would incur on a genuinely unseen specimen, and the extent of this understatement is not visible from the row-wise test score alone.

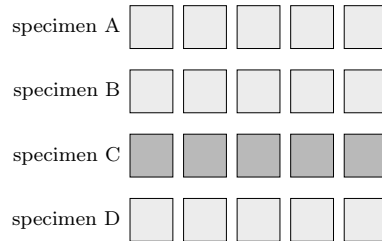
The project’s later implementation phases addressed this by grouping observations at the specimen level before splitting: every image belonging to a given specimen is assigned to the same partition, so that a specimen appearing in the test partition never also appears in training. Figure 3.1 contrasts the two approaches. This grouping principle is applied consistently, regardless of which of the three evaluation regimes described next is used to decide which specimens are held out.

(a) Naive random split of image observations



Every specimen contributes images to both the training partition (light) and the test partition (dark): a model can learn to recognise specimen-specific appearance rather than generalisable corrosion cues.

(b) Specimen-level grouped split



All images of a given specimen are assigned to the same partition: the test partition (specimen C) contains only specimens absent from training.

Figure 3.1: Report schematic contrasting a naive row-wise random split, which can place images of the same specimen on both sides of a train/test partition, against a specimen-level grouped split, which cannot. Not an experimental result.

3.3 Evaluation Regimes

Specimen-level grouping determines that whole specimens, not individual images, are the unit of assignment to a partition. It does not by itself determine *which* specimens are held out, or what question the resulting evaluation answers. The project uses three distinct regimes for this purpose, each targeting a different question, summarised in Table 3.1 and illustrated schematically in Figure 3.2.

3.3.1 Grouped specimen holdout

A random subset of specimens is held out from the pooled observed dataset without deliberately withholding an entire treatment or campaign. It therefore provides an interpolation-oriented unseen-specimen evaluation, answering the question of how well a model generalises to specimens it has not seen, drawn at random from the same pooled experimental mixture it was trained on. This is distinct from stratified sampling: because the subset is drawn at random rather than balanced by design, individual folds may be imbalanced in campaign or treatment composition and need not contain every subgroup. It is the regime closest to ordinary interpolation within the observed design space, and, where repeated evaluation is used, results are averaged over multiple random selections of the held-out subset.

3.3.2 Leave-one-treatment-out

All specimens sharing a given treatment protocol are held out together, cycling over the available treatment groups so that every group is used as the test partition exactly once. This answers a more demanding question: whether a model’s predictions transfer to a treatment protocol it has never encountered at all, rather than merely to unseen specimens of protocols it has already seen examples of. Because some treatment groups are represented by very few specimens (Table 2.1), individual leave-one-treatment-out folds can be based on small held-out samples, and results for the smallest groups are correspondingly less stable.

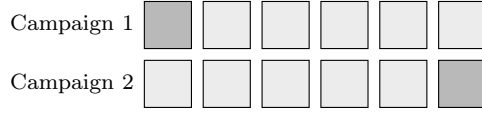
3.3.3 Leave-one-campaign-out

All specimens belonging to one full experimental campaign are held out, and the model is trained on the other campaign alone. This is the strongest distribution shift the dataset can express, since, as established in Section 2.6, campaign identity coincides with a specific combination of steel-mesh configuration, chloride concentration, and terminal ageing duration. A model evaluated under leave-one-campaign-out is therefore not being tested on specimens drawn from the training distribution; it is being tested on specimens whose design characteristics differ systematically, along several correlated dimensions at once, from every specimen available for training. For this reason, leave-one-campaign-out performance in this dataset should not be read as an estimate of expected performance on future, similarly-designed specimens; it is better understood as a stress test of extrapolation across the experimental design space, and the two readings can differ substantially. This report does not present leave-one-campaign-out results in this chapter — the regime is defined here only as a methodological tool; its results, and what they imply, are discussed once the relevant modelling activities are described in Chapters 4 and 5.

Table 3.1: The three specimen-level evaluation regimes used across the project, and the question each is designed to answer.

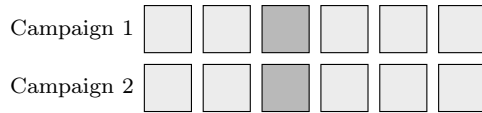
Regime		Held-out unit	Question answered
Grouped	specimen	Random subset of specimens	Generalisation to unseen specimens drawn from the same observed experimental mixture.
Leave-one-treatment-out		All specimens of one treatment protocol	Transfer to a treatment protocol not seen at all during training.
Leave-one-campaign-out		All specimens of one full campaign	Robustness under the strongest available distribution shift; partly an extrapolation test across correlated design variables (Section 2.6).

(a) Grouped specimen holdout (GroupShuffleSplit)



A random subset of specimens, drawn from across both campaigns and treatment groups, is held out. Tests generalisation to unseen specimens from the same observed experimental mixture.

(b) Leave-one-treatment-out (LOTO)



All specimens sharing one treatment protocol are held out at a time, cycling over treatment groups. Tests whether a model generalises across treatment protocols rather than memorising treatment-specific corrosion patterns.

(c) Leave-one-campaign-out (LOCO)



All specimens belonging to one full campaign are held out. Because campaign covaries with mesh count, chloride concentration, and exposure duration (§2.6), this is the strongest available distribution shift and is partly an extrapolation test rather than ordinary interpolation.

Figure 3.2: Report schematic of the three evaluation regimes, for an illustrative grid of specimens organised by campaign and treatment group. Held-out (test) specimens are shown in the darker shade. The layout is illustrative and does not reproduce the exact specimen counts of Table 2.1.

3.4 Image Representation Strategies

Across its successive phases, the project explored five broad strategies for converting an image observation into a representation usable by a predictive model:

- A. **Threshold and handcrafted colour descriptors.** A small number of scalar quantities computed from fixed colour-space thresholds (for example, the proportion of pixels falling within a rust-coloured range), used by the earliest implementation phases.
- B. **Pretrained deep embeddings.** A fixed-length vector representation obtained by passing each image through a convolutional network pretrained on a large natural-image dataset and used as a frozen feature extractor, without further training of the network itself.
- C. **Interpretable multi-family features.** A broader, explicitly engineered set of descriptors spanning colour, texture, surface morphology, and spatial location of corrosion along the specimen, designed so that each feature has a stated physical or visual interpretation.
- D. **RGB/HSV summary representations.** Colour-space summary statistics computed in both RGB and HSV representations, introduced specifically to test whether a compact, interpretable colour description adds value once specimen-design metadata is already available, in the project’s terminal-load-focused analysis.
- E. **Raw-image deep classification (planned).** Direct classification of raw images into a discrete corrosion-severity category using a deep image classifier, which is the intended representation strategy for the later four-class classification effort described in Chapter 4; at the time of writing, this strategy has been prepared for (through a dedicated, leakage-safe data and augmentation pipeline) but not yet trained or evaluated.

These strategies are introduced here only as a taxonomy; the specific feature definitions used within each strategy are not repeated in this chapter and, where needed for reference, are catalogued in an appendix.

3.5 Metadata and Multimodal Inputs

A recurring methodological question across phases is whether an image-derived representation, of any of the strategies above, contributes predictive information beyond what is already available from specimen and exposure metadata (Table 2.1) alone. This question matters because metadata and image-derived quantities are not, in general, independent sources of information in this dataset: a specimen’s treatment and campaign membership can itself be predictive of its corrosion appearance or its structural condition, independently of whatever a photograph of it shows.

The project’s treatment of this question was not uniform across phases, and the framework below describes how the comparison was eventually formalised rather than a design followed from the outset. The deep-embedding phase compared an image-only configuration against a combined image-plus-metadata configuration, but did not include a comparable metadata-only baseline alongside them. It was the later terminal-load-focused work that made an explicit three-way comparison — an *image-only* configuration using an image representation without metadata, a *metadata-only* configuration using specimen and exposure design variables without any image-derived quantity, and a *combined* (multimodal) configuration using both together — a central and consistently applied part of its methodology. Where this three-way comparison is available, it is the project’s principal tool for distinguishing predictive value that is genuinely attributable to the visual appearance of corrosion from predictive value that is instead attributable to the specimen’s known design and exposure conditions; where only the narrower two-way comparison

is available, this is stated explicitly alongside the corresponding result. The outcomes of these comparisons, for the phases in which they were carried out, are reported in Chapters 4 and 5.

3.6 Model Families

Table 3.2 lists the model families used across the project’s phases. The choice of family in each phase was pragmatic rather than exploratory: regularised linear regression provides an interpretable, low-variance baseline that is particularly informative when the training sample is small (as it is for the structural targets, Section 2.4); tree-ensemble methods (Random Forest, Gradient Boosting, XGBoost, CatBoost) provide stronger non-linear fits with built-in feature-importance diagnostics; a small multilayer perceptron was used to combine a fixed-length embedding with tabular metadata; a pretrained convolutional network provided image embeddings as described in Section 3.4; and a small family of parametric and non-parametric curve models was used specifically to fit specimen-level degradation trajectories over time (Section 3.8). This chapter does not develop the mathematical detail of any of these families; it records only their role in the project.

Table 3.2: Model families used across the project, and their role.

Family	Role in the project
Ridge / linear regression	Regularised, low-variance baseline; particularly relevant given the small number of structural observations.
Random Forest	Non-linear regression and classification baseline with feature-importance diagnostics; used across most phases and targets.
Gradient Boosting	Alternative non-linear ensemble, compared against Random Forest per task.
XGBoost	Alternative gradient-boosted ensemble, benchmarked in the earliest classical phase and again, alongside other candidate families, in the later terminal-load-focused modelling.
CatBoost	Additional gradient-boosted ensemble candidate introduced in later structural-condition modelling; used with the same numeric feature tables as the other ensemble methods, not configured to use its native categorical-feature handling.
Multilayer perceptron (MLP)	Combines a fixed-length image embedding with tabular metadata; also used as an image-only regressor for comparison.
Pretrained convolutional network	Frozen (not fine-tuned) feature extractor producing the fixed-length image embedding consumed by the MLP above.
Degradation-curve models	A small family of parametric (linear, logarithmic, logistic, Gompertz) and non-parametric (monotone-isotonic) curves fitted to estimated condition over the observation period, as input to the proxy-RUL computation (Section 3.8).

3.7 Evaluation Metrics

Model performance is reported using a small, consistent set of metrics, defined here so that later chapters can use them without redefinition. For regression targets, the project reports the mean absolute error (MAE) and root-mean-square error (RMSE), both in the physical units of the target; the coefficient of determination (R^2); and the Spearman rank correlation between predicted and observed values. For the four-class classification target, the project’s planned evaluation (not yet carried out; Section 3.4, strategy E) specifies overall accuracy, macro-averaged F1 score, and per-class recall reported through a confusion matrix.

Spearman rank correlation is reported alongside the absolute-error metrics because it evaluates whether a model correctly orders specimens by severity without requiring its predictions to be correctly scaled or calibrated in absolute terms, and because it is markedly less sensitive than MAE or RMSE to a small number of specimens with large absolute errors. A model can be poorly calibrated in absolute magnitude while still correctly identifying which specimens are more severely affected than others, and the two metrics together distinguish this situation from one in which a model’s ranking of specimens is itself unreliable.

Macro-averaged F1, rather than overall accuracy, is identified in advance as the primary metric for the planned four-class classification task because the four severity classes are markedly imbalanced in the underlying data (Table 2.2): the lowest-severity class accounts for the large majority of observations. A classifier that predicted the majority class for every observation would achieve high overall accuracy while providing no discrimination among the minority, higher-severity classes that are of greatest practical interest. Macro-averaged F1 weights each class equally regardless of its frequency and is therefore markedly more informative than accuracy alone for this task; per-class recall and the confusion matrix are reported alongside it to characterise which classes, specifically, a classifier confuses.

3.8 True Remaining-Useful-Life versus Proxy-RUL

This section states explicitly a methodological boundary that governs every subsequent discussion of service-life or degradation in this report.

The dataset, as established in Section 2.4, does not contain the supervision required to train a model to predict remaining useful life, or time to failure, in the ordinary supervised sense. Such a target would require either a recorded failure event time for each specimen or repeated structural measurements of the same specimen at multiple points in time, so that the rate and trajectory of structural degradation could be observed directly. Neither is present: each specimen contributes a single terminal structural measurement, taken once, at the end of its observation period, and no specimen is observed to fail during a monitored period with the failure time recorded.

In place of a directly supervised target, later project phases constructed a screening-oriented proxy through an explicit, multi-step derivation:

1. A model trained on the available (dense, visible-corrosion or metadata-linked) supervision produces an *estimated* structural or corrosion condition at each observation week for a given specimen, including weeks at which no direct measurement exists.
2. These estimated, per-week values are used to fit a specimen-level degradation curve over the observation period, using one of the curve families listed in Table 3.2.
3. An engineering threshold is defined on the quantity described by that curve (for example, a specified level of estimated wire-area loss).
4. The time at which the fitted curve is projected to cross that threshold, measured forward from the specimen’s last available observation, is reported as the proxy remaining-useful-life.

Every step in this derivation is model-dependent: it depends on the accuracy of the condition estimates produced in step 1, on the curve family selected in step 2, and on the threshold chosen in step 3. None of these choices is fixed by direct observation. The resulting quantity is therefore an exploratory, screening-oriented output, useful for prioritising which specimens might warrant closer inspection, and it is not, and should not be presented as, an observed or directly supervised lifetime measurement. This report consistently uses the qualified term *proxy-RUL* for this quantity and reserves *remaining useful life* or *RUL* without qualification for a genuinely observed lifetime target, which this dataset does not provide.

3.9 Methodological Principles Carried Forward

Not every phase of the project met the standards described in this chapter: the earliest exploratory prototype, in particular, did not evaluate its production model against any held-out partition at all, and Chapter 4 reports this and other historical shortcomings where they occurred rather than treating this chapter’s framework as retroactively true of every prior phase. The principles below instead describe the position reached by the project’s mature pipeline — the implementation phases responsible for the majority of the results discussed in Chapters 4 and 5 — and are adopted in this report as the standard against which a result is judged to constitute valid evidence. Where an earlier or partial phase departs from one of these principles, that departure is noted explicitly at the point the result is discussed, rather than silently overlooked or silently corrected:

- Evaluation is grouped at the specimen level; a result is not treated as valid evidence of generalisation if it comes from a split that places images of the same specimen in both training and test partitions.
- Wherever an image-derived representation is proposed as predictive of an outcome, its contribution is, from the terminal-load-focused phase onward, checked against an explicit image-only, metadata-only, and combined comparison; earlier phases that instead compared only two of these three configurations are identified as such (Section 3.5).
- Model robustness is preferably assessed under more than one evaluation regime, including at least one regime designed to expose distribution shift (leave-one-treatment-out or leave-one-campaign-out), rather than reported under a single grouped holdout alone.
- Observed quantities, model-derived quantities, and derived-screening quantities (Table 2.3) are kept terminologically distinct throughout this report, and a derived-screening output is never described in language implying it is a direct measurement or a supervised prediction of an observed target.
- Proxy-RUL outputs are interpreted conservatively, as exploratory screening indicators contingent on the modelling choices used to derive them, not as validated lifetime estimates.
- Where a directly measured endpoint is available for a quantity of interest, the project’s later work favours modelling that endpoint directly over constructing a further derived proxy for it.

Chapter 4

Research and Development Activities

This chapter reconstructs the project as a chronological scientific and engineering investigation. Each section addresses one implementation phase in turn, but the organising logic is not "what was built" so much as: what question the phase was trying to answer, what the resulting evidence showed, what limitation that evidence exposed, and what decision followed from it. Provenance labels for each phase are given only in footnotes and figure/table captions, where they matter for traceability; the prose refers to phases by their scientific role. The shared methodological vocabulary — specimen-level grouping, the three evaluation regimes, the representation-strategy taxonomy, and the distinction between observed, model-derived, and derived-screening quantities — was established in Chapter 3 and is used here without re-derivation.

4.1 Exploratory Prototype

The project's first implementation was an informal exploratory prototype, built to establish whether the available photographs contained any usable signal at all rather than to produce a validated model.¹ It computed a single fixed-threshold colour mask per image, derived a handful of additional descriptors by parsing the specimen identifier as a string (rather than reading the structured metadata table), and trained a RandomForest to predict next-week visible rust percentage. A separate script then applied this one-step regressor autoregressively — repeatedly feeding each week's prediction back in as the next week's input — until a fixed 10% rust threshold was crossed, and reported the number of simulated steps as a "remaining useful life."

This should not be read as a trained or validated RUL model: it is a heuristic simulation built on a single-step regressor, compounding that regressor's error at every simulated step, with no mechanism to check the simulated trajectory against an observed one. More consequentially, the script that produces the prototype's "production" model fits on the entire available dataset with no held-out partition at all, and no metrics from this phase were saved anywhere in the repository — every quantitative claim about this phase's performance would have to come from console output that no longer exists. The phase is therefore useful only as a starting point: it suggested that a colour-based signal was worth pursuing, while simultaneously demonstrating that a benchmarked, leakage-aware, and reproducible pipeline did not yet exist. That gap is what the next phase was built to close.

4.2 Classical Machine-Learning Baseline

The second phase replaced the prototype with a benchmarked pipeline: a consistent set of candidate regressors (ElasticNet, RandomForest, Gradient Boosting, and XGBoost) evaluated

¹Corresponds to the earliest exploratory implementation referenced in Chapter 2's corrupted-image-handling table.

under a specimen-grouped holdout followed by grouped cross-validation on the training partition, with metrics and feature importances persisted for every candidate. Rather than posing one prediction problem, the phase implemented three distinct tasks over the same underlying data, distinguished here because they have different scientific meanings and must not be merged.

Current visible-corrosion estimation predicted the current-week peak rust percentage from a combination of handcrafted image descriptors and specimen/exposure metadata. The selected RandomForest achieved a grouped holdout MAE of 2.38 and $R^2 = 0.900$ — the strongest headline number produced by this phase. Inspecting the model’s own feature importances, however, shows that a single handcrafted feature — the percentage of pixels falling inside a fixed rust-colour threshold — accounts for over 90% of the total importance. Because this same colour threshold is also how the target label itself is effectively derived, the result is close to a restatement of the labelling rule rather than independent evidence that the model has learned a generalisable visual concept of corrosion. The headline accuracy is real, but it does not by itself establish that the model has captured visual information beyond the threshold used to construct its own label.

Progression modelling predicted the same current-week rust target from specimen/exposure metadata and observation week alone, with no image input at all. This is a metadata- and week-conditioned regression, not a sequence model: it does not consume a specimen’s own history of prior observations and does not forecast forward from one time step to the next. The selected Gradient Boosting model reached a grouped holdout R^2 of only 0.517, substantially weaker than the image-assisted task above, and the result should be read accordingly — as an estimate of how much predictive information design and exposure variables alone carry for this target, not as a temporal forecasting result.

Threshold-timing estimation predicted the number of weeks remaining until peak rust crosses a fixed 20% threshold, using the specimen’s true current corrosion level (not a model’s own prediction of it) together with metadata as inputs. The selected RandomForest reached a grouped holdout MAE of 2.21 weeks. This is a derived timing quantity conditioned on an observed covariate, not an instance of remaining-useful-life prediction in the sense defined in Section 3.8: it times a corrosion threshold, not a structural failure or service-life endpoint, and it is trained on already known rather than forecast inputs.

Table 4.1: Representative grouped-holdout results, classical baseline phase. Selected model and holdout metrics as saved by the phase’s own evaluation script.

Task	Selected model	MAE	RMSE	R^2
Current corrosion (image + metadata)	RandomForest	2.38	4.25	0.900
Progression (metadata + week only)	Gradient Boosting	6.71	9.33	0.517
Threshold-timing (weeks to 20%)	RandomForest	2.21 wk	3.09 wk	0.782

Two further findings from this phase materially affect how its results should be read. First, the specimen-grouped holdout used throughout this phase was a genuine improvement over the exploratory prototype’s absence of any held-out evaluation, and is the same grouping principle formalised in Section 3.2. Second, model selection within each task was based on the holdout metric itself rather than on a separate validation partition: for the threshold-timing task, an alternative candidate achieves a better holdout RMSE (2.82 vs. 3.09) and R^2 (0.818 vs. 0.782) than the selected RandomForest, which is chosen because its holdout MAE is marginally lower (2.21 vs. 2.24). This is a historical model-selection practice specific to this phase, not the methodological standard adopted later in the project (Section 3.9), and it means the reported holdout

numbers should be read as an optimistic upper bound on this phase’s true generalisation performance rather than as an untouched final estimate. The pipeline’s reproducibility is also limited in the current repository state: re-running it against the presently stored data workbook raises a schema mismatch, so the numbers above are the saved record of a historical run rather than a result that can be freshly regenerated without repair.

Taken together, the phase established that a benchmarked, specimen-grouped pipeline could produce strong-looking results for current-corrosion estimation, but left open whether that strength reflected a genuinely useful image representation or simply a restatement of the handcrafted threshold used to build the label. Testing whether a representation learned directly from image content, rather than a fixed hand-designed threshold, could resolve this question motivated the next phase.

4.3 Deep Image-Embedding Experiment

The third phase replaced the handcrafted colour threshold with a representation learned by a convolutional network: each image was passed through a pretrained ImageNet ResNet-18 with its final classification layer removed, used purely as a frozen feature extractor (no weights in the backbone were updated), producing a 512-dimensional embedding. The experiment targeted the same family of quantity as the previous phase’s strongest result — current-week peak rust percentage — so that the value of a learned representation could be assessed against a comparable target, computed here on the phase’s own 792-row dataset basis, with the one corrupted image’s embedding replaced by the column-wise median.

Two configurations were compared: an image-only regressor consuming only the 512-dimensional embedding, and a multimodal regressor consuming the same embedding concatenated with seven metadata fields. No metadata-only baseline was trained in this phase, so the comparison that later became central to the project’s methodology (Section 3.5) is only partially available here.

The results contain a genuine reversal worth stating clearly. Across training, the multimodal model reached a lower best validation MAE than the image-only model (4.02 vs. 4.96). On the held-out test partition, however, the ranking inverted: the image-only model achieved a lower test MAE than the multimodal model (5.63 vs. 5.75). Figure 4.1 shows both comparisons side by side. The phase’s own pipeline selected the final deployed model by test MAE — the same partition used to report the final metric — so the image-only model was deployed on the strength of a test-set comparison that also served as the reported result, rather than being confirmed by an independent partition after selection. This mirrors the model-selection pattern already noted in Section 4.2 and should likewise be read as a historical practice specific to this phase, not a result that establishes multimodal fusion as inferior in general.

A second finding concerns where the deployed model’s errors concentrate. Grouping the saved test-set residuals by observation week shows a systematic pattern: mean absolute error rises from 2.22 at week 10 or earlier, to 6.62 across weeks 12–20, to 12.08 from week 23 onward, and the average bias moves from a small positive value (+0.85) to a substantial negative one (−6.91) over the same progression. Figure 4.2 shows this directly: the model increasingly underestimates severity as corrosion advances, rather than erring in either direction at random.

These findings support a narrower conclusion than either "deep embeddings work" or "deep embeddings failed." A frozen deep representation was feasible for current-week visible-corrosion estimation, reaching accuracy broadly comparable to the handcrafted-feature baseline. Adding metadata to that representation did not yield a stable improvement in this saved experiment: it helped during training but not on the final held-out partition, and the comparison was not confirmed independently of the selection step. The representation also offered materially less interpretability than the explicit handcrafted descriptors of the previous phase — a 512-dimensional embedding does not, by inspection, reveal which visual cues drive a prediction the way a rust-percentage threshold does. Combined with the late-stage underestimation problem and the ab-

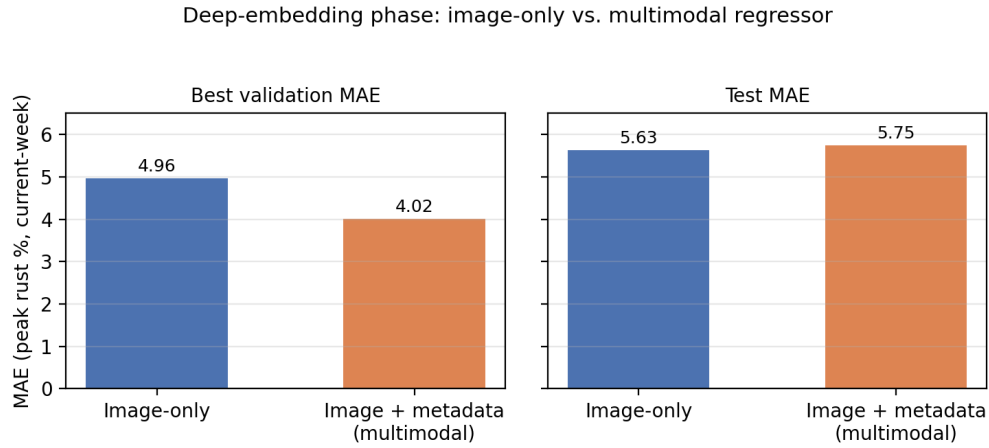


Figure 4.1: Deep-embedding phase: best validation MAE versus test MAE for the image-only and multimodal (image + metadata) regressors, current-week peak rust percentage. The multimodal model is ahead on validation but behind on test; the deployed model was chosen by the test-partition comparison shown on the right. Generated from the phase’s own saved training history and model-comparison tables.

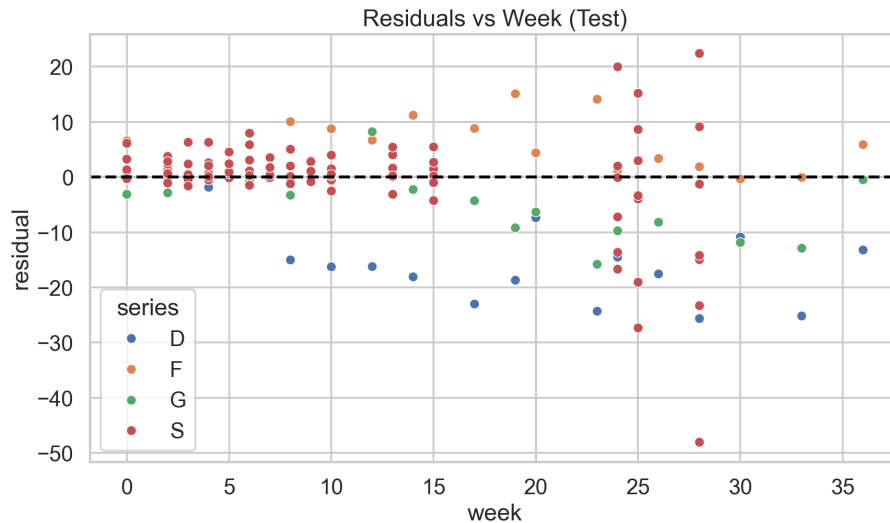


Figure 4.2: Deep-embedding phase, image-only model: residuals (predicted minus observed peak rust percentage) on the held-out test partition, by observation week and specimen series. Residuals drift increasingly negative at later weeks, indicating systematic underestimation of severe, late-stage corrosion. Reproduced from the phase’s own saved diagnostic figure.

sence of any progress on structural or remaining-service-life questions, these results motivated a return to explicit, interpretable image descriptors, alongside a shift toward posing more demanding questions about hidden structural condition rather than continuing to refine current-corrosion estimation alone.

4.4 Interpretable Corrosion Quantification and Structural-Condition Feasibility

The fourth phase is the project’s largest single body of work in this period and marks a scientific turning point. It replaced the deep embedding with a broad, explicitly engineered feature representation — ninety-one descriptors spanning corrosion amount, spatial distribution along the specimen, surface morphology, colour in three colour spaces (RGB, HSV, and Lab), and texture — computed by a Python image-processing pipeline. This pipeline was inspired by the predecessor thesis’s manual image-analysis workflow but is best described as a Python reconstruction and extension of that workflow’s ideas rather than an exact reproduction of it: the original workflow combined manual GIMP/BIMP editing steps with a separate MATLAB analysis stage, whereas this phase implements comparable descriptors programmatically, in some cases extending beyond what the manual workflow computed.

This phase was also the first to pose two further questions beyond current-corrosion estimation: whether visible corrosion could help estimate a specimen’s hidden structural condition, and whether a degradation trajectory and threshold-crossing screening quantity could be derived from the resulting estimates. Both required confronting the asymmetry between dense visible-corrosion supervision and sparse structural supervision established in Section 2.4, and both were evaluated under all three of the grouped evaluation regimes defined in Section 3.3 — the first phase in the project to do so systematically.

4.4.1 Visible-corrosion modelling

For the two continuous visible-corrosion targets, several candidate model families were compared (linear regression, RandomForest, histogram-based gradient boosting, and a small multilayer perceptron); the specific family selected as best differs by target and by evaluation regime, and is not repeated model-by-model here. Two categorical severity targets were also modelled, using the historical *five*-category ordinal scheme in place at this phase’s dataset snapshot (Section 2.5) — these results must not be described as four-class classification, which belongs to the later, separate dataset and classification effort (Section 3.4, strategy E).

Table 4.2 reports the best mean result for each target under each evaluation regime. Under a grouped specimen holdout, both continuous targets are estimated with useful accuracy (total rust: MAE 1.38, $R^2 = 0.646$; peak rust: MAE 4.21, $R^2 = 0.784$). Robustness to campaign holdout is reasonable — accuracy declines but both targets retain positive R^2 under leave-one-campaign-out. Robustness to treatment holdout is markedly weaker: mean R^2 across treatment folds becomes negative for both targets, reflecting that individual treatment groups are small (Section 3.3.2) and that at least one held-out treatment fold produces a strongly negative per-fold R^2 , pulling the average down. The five-category classification targets follow a similar pattern: grouped holdout accuracy is moderate to good (0.83 for total-rust category, though macro-F1 of 0.573 reveals the imbalance-sensitive weakness that accuracy alone would hide), and macro-F1 does not exceed roughly 0.47 under either stricter regime. Read together, visible-corrosion estimation is the most robustly image-driven layer of this phase: it degrades under distribution shift, as expected of any model facing genuinely new treatment or campaign conditions, but it does not collapse.

4.4.2 Structural-condition feasibility

The structural-condition models require more careful description, because several aspects of their design materially affect what the results can be taken to show. As established in Section 2.4, only 48 rows in the dataset carry a structural label, one per specimen. The feature set used to predict these targets is not image-only: it combines specimen/exposure metadata, the *manually assigned* visible-corrosion labels and their categories (the ground-truth annotations, not the previous subsection’s model predictions of them), and a subset of the engineered image descriptors. The pipeline’s own predicted corrosion values are not included in this feature list, so the damage stage does not chain a learned corrosion prediction into a learned damage prediction; it instead poses a single, direct estimation problem — from metadata, manual labels, and image descriptors together to structural condition — rather than a strictly image-based inference.

Under a grouped specimen holdout, both structural targets show a modest but non-trivial relationship to the available features: wire-area loss reaches MAE 3.90 ($R^2 = 0.310$), and ultimate load reaches MAE 0.121 kN ($R^2 = 0.385$) on this phase’s own 38/10 grouped-holdout split. These are this phase’s own historical results, computed on its own 792-row dataset basis and its own feature engineering, and are not compared numerically against the later terminal-load-focused results discussed in a subsequent section, which use a different pipeline, feature set, and dataset alignment.

Under either stricter evaluation regime, performance deteriorates substantially. Leave-one-campaign-out mean R^2 becomes negative for both targets (-0.39 for wire-area loss, -8.36 for ultimate load), and leave-one-treatment-out is similarly negative and, for ultimate load, more severely so (-4.57). Figure 4.3 places these alongside the visible-corrosion results from the previous subsection: the contrast between the two rows of bars is the central diagnostic finding of this phase. The available evidence does not support reliable image-based recovery of hidden structural condition under either stricter regime tested here. The visible-corrosion layer is markedly more robust overall — both of its targets retain positive R^2 under leave-one-campaign-out — but it is not uniformly robust: leave-one-treatment-out is less stable and drives both visible-corrosion targets to negative mean R^2 as well (Table 4.2), so the two stricter regimes should not be treated as interchangeable when comparing the two target families.

Three further diagnostics, reported directly in this phase’s own technical report, explain why. First, correlation between the manually labelled visible-corrosion measures and wire-area loss is close to zero across the 48 structural rows (total rust: $r = 0.021$; peak rust: $r = 0.011$) — visible corrosion, as measured here, is only weakly related to internal wire loss even before any modelling is attempted. Second, feature-importance analysis of the full-data ultimate-load model shows its largest contributions coming from steel-mesh count and campaign indicator variables, not from image descriptors — the same campaign-linked metadata dominance later quantified more thoroughly in Section 2.6’s discussion of design confounding. Third, the "final" structural estimator for each target is fit on all 48 structural rows with no further held-out partition at that step, and is then applied to every observation in the dataset — including non-terminal weeks — to produce an estimated structural condition at each week. These estimates are the model-derived quantities introduced in Table 2.3: they are the output of a model fit in-sample on the full structural subset, not a repeated physical measurement, and the degradation curves built from them in later work inherit this property.

4.4.3 What this phase established

This phase’s evidence supports several specific conclusions, stated here without extrapolation beyond what was tested. Interpretable image descriptors can estimate visible corrosion with useful accuracy, including under a meaningful campaign-holdout stress test. Visible-corrosion estimation is considerably better supported by the available evidence than hidden structural-condition estimation. Structural inference in this dataset is constrained by the combination of

Table 4.2: Representative results by target and evaluation regime, interpretable multi-family feature phase. Model shown is the best-mean-MAE candidate for that (target, regime) cell. Rust and wire-area-loss targets are in percent; ultimate load is in kN. All results computed on this phase’s own 792-row dataset basis; see Section 2.5 for how this differs from the later canonical basis.

Target	Regime	Model	MAE	R^2
Total rust (visible)	Grouped hold-out	RF	1.38	0.646
Total rust (visible)	LOTO	MLP	1.53	−0.677
Total rust (visible)	LOCO	RF	2.10	0.582
Peak rust (visible)	Grouped hold-out	MLP	4.21	0.784
Peak rust (visible)	LOTO	RF	4.74	−0.077
Peak rust (visible)	LOCO	HGB	6.21	0.595
Wire-area loss (structural)	Grouped hold-out	RF	3.90	0.310
Wire-area loss (structural)	LOTO	HGB	10.76	−5.481
Wire-area loss (structural)	LOCO	HGB	12.28	−0.387
Ultimate load (structural)	Grouped hold-out	RF	0.121	0.385
Ultimate load (structural)	LOTO	RF	0.158	−4.569
Ultimate load (structural)	LOCO	ExtraTrees	0.624	−8.356

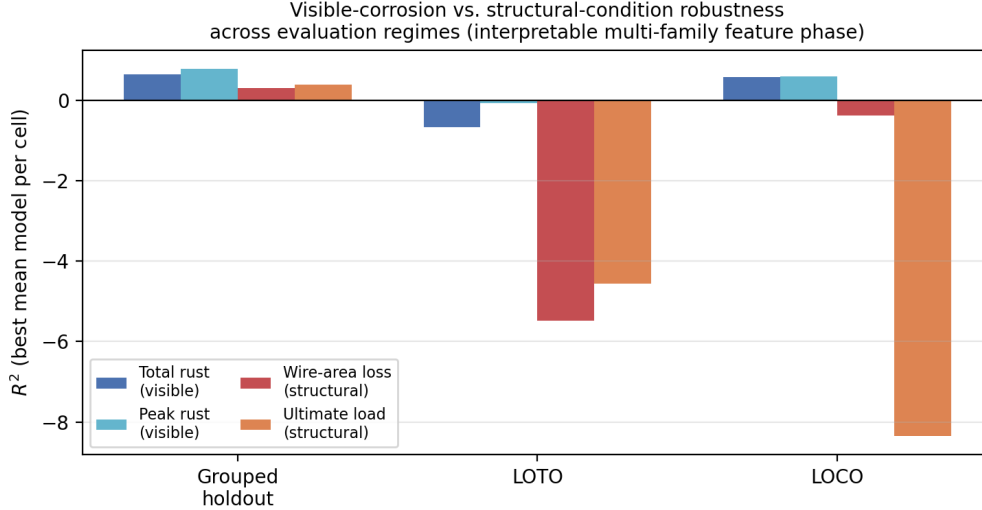


Figure 4.3: Best-mean-model R^2 by evaluation regime for the two visible-corrosion targets and the two structural-condition targets, interpretable multi-family feature phase. Visible-corrosion models are substantially more robust than the structural models overall, and both visible-corrosion targets retain positive R^2 under leave-one-campaign-out; leave-one-treatment-out is less stable and produces negative mean R^2 for both visible-corrosion targets as well, while the structural targets deteriorate far more severely under both stricter regimes. Generated directly from the phase’s own saved regression-metrics tables (Table 4.2).

sparse supervision (48 rows, one per specimen), small and unevenly sized treatment groups, and campaign-linked design variables that dominate the structural models’ feature importances. The structural-condition stage is a mixed-input estimation problem — metadata, manual corrosion labels, and image descriptors together — not an image-only inference.

It is also this phase’s own technical report, not a subsequent phase, that first names campaign confounding and the risk of a "downstream label shortcut" (structural models drawing signal from manually assigned corrosion labels and design metadata rather than from independently learned image content) as explicit concerns. Work described in later sections that quantifies these problems more rigorously and responds to them with dedicated diagnostics and a reframed research question should therefore be understood as acting on a problem this phase had already identified, not as discovering it for the first time.

That reframing followed directly from the findings above. The sparse, campaign-confounded structural signal recovered here motivated stronger attention to out-of-sample structural estimation, explicit image-versus-metadata ablations, robustness-oriented model selection, feature cleanup, and monotonicity-aware degradation modelling, and ultimately a reconsidered structural research question targeting one of the two directly measured terminal endpoints available in the dataset.

4.5 Degradation Modelling and Proxy-RUL Feasibility

The same phase also carried out a feasibility experiment in the derived screening quantity introduced in Section 3.8: for each of the 48 specimens, a degradation curve was fitted to each of four per-week series — the two directly observed visible-corrosion measures and the two model-estimated structural measures from Section 4.4.2 — and projected forward to identify when it would cross an engineering threshold. This section reports what that experiment actually produced, and is deliberately brief: the exercise is a feasibility check on the derivation chain itself, not a second opportunity to validate structural prediction.

Curve fitting used piecewise-linear and exponential families, selected per specimen and target. Fit quality was good for the two visible-corrosion series (median R^2 of 0.90 and 0.92) and moderate for the two model-estimated structural series (median R^2 of 0.64 and 0.60) — every curve was fit to the same 15–18 observation weeks per specimen established in Section 2.3, so the structural curves are trajectories through model-estimated values, not through repeated physical measurements. The final proxy-RUL for a specimen was defined as the earliest of three threshold-crossing events read off these curves and projected up to 104 weeks beyond the last observation: a 25% wire-area-loss crossing, an ultimate-load crossing at 80% of a campaign reference load, or a crossing of a composite health index combining all four series.

Of the 48 specimens, 33 received a finite proxy-RUL estimate within the 104-week projection horizon; the remaining 15 never crossed any threshold within that horizon and are right-censored. Of the 33 finite estimates, 23 — the large majority — are exactly zero weeks, meaning the relevant threshold was already crossed at the specimen’s last available observation rather than projected into the future. Figure 4.4 shows this concentration directly. Only the remaining ten specimens carry a proxy-RUL that is both finite and genuinely forward-looking, ranging up to 55.5 weeks.

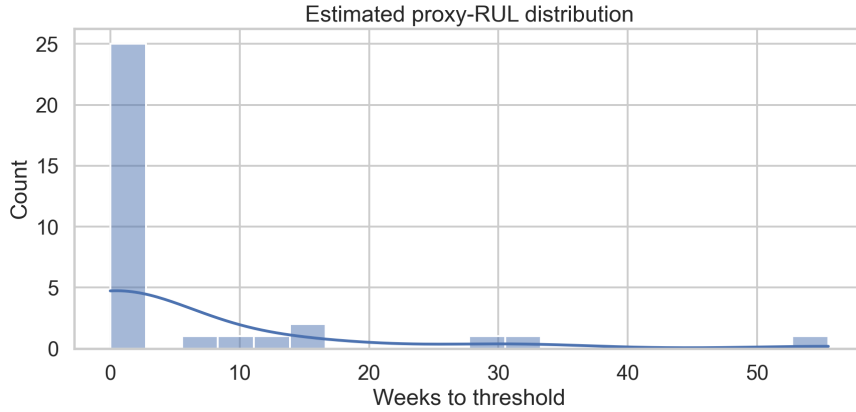


Figure 4.4: Distribution of finite proxy-RUL estimates (33 of 48 specimens); the remaining 15 specimens are right-censored (no threshold crossing within the 104-week projection horizon) and are not shown on this axis. Most finite estimates are at or near zero weeks. Reproduced from the phase’s own saved output.

This outcome illustrates, concretely, why the quantity is a feasibility result rather than a validated remaining-service-life estimate. Every stage of the derivation compounds the uncertainty of the stage before it: the structural curves are fitted to estimates that are themselves weak under distribution shift (Section 4.4.2), the curve family and threshold are modelling choices rather than physical constants (Section 3.8), and the 104-week extrapolation horizon extends well beyond any observed data for most specimens. The pipeline is a working demonstration that the four-step derivation from repeated observed and model-estimated condition series to a threshold-crossing screening quantity can be executed end to end; it does not, and was not designed to, constitute evidence that the resulting numbers are accurate estimates of true remaining service life.

4.6 Robustness and Diagnostic Refinement

The next phase of work continued directly from the structural-condition findings above, using the same underlying dataset structure but a more rigorous implementation. This phase’s contribution should be stated precisely: campaign confounding, sparse structural supervision, and the risk of models drawing on manually assigned labels rather than independently learned image content were all already named, in those terms, by the previous phase’s own report. This phase’s

contribution was to quantify how severe these problems were under systematic testing, audit the feature set for redundancy, and revise the modelling and selection procedure in response — acting on a diagnosis the project had already made, not making it for the first time.

A feature-redundancy audit identified thirteen near-constant columns, one exact duplicate pair, and thirty-six feature pairs with rank correlation of 0.95 or higher within the engineered feature set — including a pairing already anticipated in Section 4.4.1, since one image descriptor and the total-rust label are correlated at $\rho = 0.9996$, a near-tautological relationship. More informative for the structural question is where genuine, non-redundant signal was found: the strongest univariate correlate of wire-area loss is a single image morphology feature, but only at $r = 0.38$; for ultimate load, the five strongest univariate correlates are not image features at all, but five metadata and temporal variables — steel-mesh count, chloride concentration, ageing days, observation week, and terminal week — each correlated with ultimate load at exactly the same magnitude, $|r| = 0.827$, a direct numerical signature of the fact that these variables are perfectly aligned with campaign identity on the 48-row structural subset (Section 2.6). This answers, for the ultimate-load target specifically, where the apparently predictive structural signal actually originates: from experimental design metadata that happens to be campaign-linked, not from the corrosion images.

The phase’s own robustness benchmark, summarised in Table 4.3 and Figure 4.5, quantifies the consequence for the two target families side by side — deliberately, because the two families behave very differently and must not be described with a single verdict. Visible-corrosion targets remain stable or even improve in relative terms under leave-one-campaign-out (surface total rust: $0.77\times$ its own grouped-holdout MAE; peak rust: $0.96\times$), with Spearman correlation staying above 0.97 in both cases. Structural targets do not: ultimate load’s leave-one-campaign-out MAE is 3.24 times its grouped-holdout value, with Spearman falling to 0.35, and wire-area loss’s leave-one-campaign-out Spearman turns negative (-0.089). The robustness collapse documented here belongs to the structural stage; it is not a property of the pipeline as a whole, and the visible-corrosion layer’s continued robustness is as much a part of this phase’s evidence as the structural collapse is.

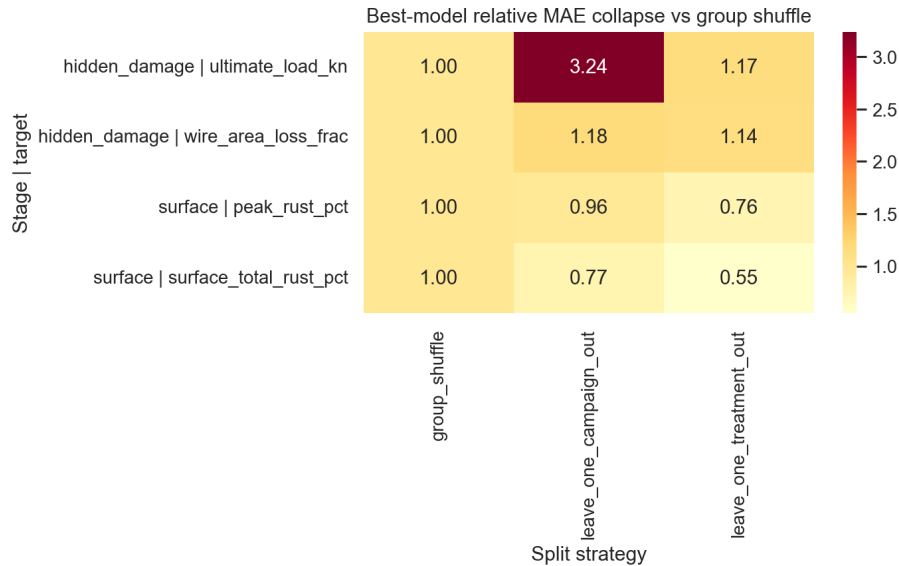


Figure 4.5: Best-model MAE under each evaluation regime, relative to that target’s own grouped-holdout MAE (1.00 by definition). Visible-corrosion targets (bottom two rows) stay at or below $1.2\times$ under every regime; structural targets (top two rows) inflate sharply under leave-one-campaign-out, most severely for ultimate load ($3.24\times$). Reproduced from the phase’s own saved diagnostic output.

Table 4.3: Grouped-holdout and leave-one-campaign-out MAE and Spearman correlation for the best-performing model per target under each individual evaluation strategy considered on its own, robustness-quantification phase. This is a per-strategy diagnostic comparison, not necessarily the phase’s final robustness-weighted model selection (see following paragraph). Rust targets in percent, wire-area loss as a fraction, ultimate load in kN. Source: the phase’s own saved robustness-benchmark table.

Target	Regime	Model	MAE	Spearman
Surface total rust (visible)	Grouped hold-out	GB	0.177	0.999
Surface total rust (visible)	LOCO	GB	0.136	0.997
Peak rust (visible)	Grouped hold-out	RF	1.097	0.992
Peak rust (visible)	LOCO	RF	1.058	0.977
Wire-area-loss fraction (structural)	Grouped hold-out	RF	0.105	0.280
Wire-area-loss fraction (structural)	LOCO	RF	0.124	−0.089
Ultimate load, kN (structural)	Grouped hold-out	RF	0.174	0.778
Ultimate load, kN (structural)	LOCO	RF	0.564	0.353

Table 4.3 should be read as a diagnostic comparison, not as a statement of which model the phase ultimately selected: it reports, independently for each target, whichever model performed best under a given evaluation strategy considered on its own. The phase’s final structural-model selection instead uses the robustness-weighted procedure described below and is recorded separately in its own saved best-model table. For ultimate load the two coincide: RandomForest on the metadata-only feature set is both the per-strategy diagnostic winner shown above and the phase’s final robustness-weighted selection, with identical metrics. For wire-area-loss fraction they differ: the per-strategy diagnostic winner in Table 4.3 is RandomForest, but the phase’s final robustness-weighted selection is CatBoost on the metadata-only feature set (grouped-holdout MAE 0.124, leave-one-treatment-out MAE 0.116, leave-one-campaign-out MAE 0.121, grouped-holdout Spearman 0.289) — numerically close to, but not identical with, the diagnostic table’s figures for that target. Both comparisons agree on the point that matters for this section: regardless of which specific model is chosen, the structural targets’ feature importances and robustness pattern remain metadata-dominant, in contrast to the visible-corrosion targets.

Two further changes distinguish this phase’s methodology from the previous one. Structural-target model selection moved from picking whichever model happened to have the best mean score on a single split, to a weighted combination emphasising leave-one-campaign-out performance most heavily — a deliberate response to the previous phase’s own observation that grouped-holdout performance alone was an unreliable guide to structural robustness. Degradation-curve fitting was also revised: the previous phase’s fits were dominated by a flexible, non-parametric curve family selected for 44 of 48 specimens, which the project’s own review noted can reach near-zero fitting error by interpolation rather than by capturing a genuine trend; the revised fitting is more evenly split between that family and a simpler linear one (21 versus 27 specimens), and degradation status is now reported as an explicit crossed/not-yet-crossed/censored category breakdown rather than a single ambiguous "weeks remaining" figure. The phase’s own stated

conclusion, after these revisions, is worth reporting directly: the improved structural models remain metadata-dominant, and the project does not consider this sufficient support for a claim that surface image features robustly infer hidden structural damage.

4.7 Refocused Terminal Ultimate-Load Study

The robustness findings above motivated a deliberate change in research question, not merely a further modelling attempt. Rather than continuing to target a sparse, model-derived, campaign-confounded hidden-damage estimate, this phase targets one of the two structural quantities in the dataset that are directly measured at a single, well-defined terminal time point: ultimate load, selected here in preference to the other (wire-area loss) for this refocused analysis. The question posed is correspondingly specific: after controlling for specimen design and exposure characteristics, does superficial visible corrosion provide predictive information about terminal ultimate load beyond what those design characteristics already carry? Because this question is about incremental value, not standalone accuracy, every analysis in this phase compares matched feature-set configurations under the same model-selection and evaluation procedure, rather than reporting a single "best" pipeline in isolation.

4.7.1 Does corrosion add value over metadata, pooled?

Table 4.4 reports the pooled, grouped cross-validation comparison across seven feature-set configurations, all 48 specimens, ten repeated grouped splits. A metadata-only Ridge regression — eleven features describing specimen design and exposure, no image content at all — achieves the lowest mean absolute error (0.173 kN) and the second-highest Spearman correlation (0.758) of any configuration tested. Adding HSV colour descriptors to the metadata leaves MAE essentially unchanged (0.173 kN) while marginally improving Spearman (0.771); adding RGB descriptors instead makes both metrics worse (MAE 0.184 kN, $R^2 = 0.584$ versus metadata-only's 0.645). Image descriptors used alone, without metadata, are clearly the weakest configurations tested (RGB-only MAE 0.253 kN, $R^2 = 0.269$). Read together, the pooled evidence does not show image descriptors adding meaningful value over metadata alone, and in one configuration (RGB) adding them measurably hurts performance.

Table 4.4: Pooled feature-set comparison, terminal ultimate load, grouped cross-validation (48 specimens, 10 splits). Model shown is the winning model for that feature set. Source: the phase's own saved comparison table.

Feature set	Features	Model	MAE	R^2	Spearman
Metadata only	11	Ridge	0.173	0.645	0.758
Metadata + HSV	47	Ridge	0.173	0.635	0.771
Metadata + RGB + HSV	88	CatBoost	0.183	0.607	0.753
Metadata + RGB	52	Ridge	0.184	0.584	0.748
HSV only	36	CatBoost	0.210	0.481	0.705
RGB + HSV	77	CatBoost	0.234	0.411	0.668
RGB only	41	CatBoost	0.253	0.269	0.501

A subset analysis restricted to the 43 specimens whose terminal corrosion was above a near-zero onset threshold shows a modest, narrower reversal: there, the combined metadata-plus-RGB-plus-HSV configuration edges ahead of metadata-only on MAE (0.163 vs. 0.173 kN) and R^2 (0.660 vs. 0.633), although metadata-only retains a marginally higher Spearman correlation (0.803 vs. 0.791) even in this subset. This is reported as a narrow, subset-specific observa-

tion consistent with image descriptors being more competitive once a specimen has developed measurable corrosion, not as a reversal of the pooled conclusion above.

4.7.2 What happens within a single mesh family?

The specimen design matrix in Section 2.1 means that campaign identity, steel-mesh count, and chloride level co-vary almost perfectly. Table 4.4 is therefore a pooled result across two structurally different specimen designs, and the phase’s own analysis goes on to ask what the same comparison shows within each mesh family alone. The answer is an important caution against reading the pooled MAE values too optimistically: within either the 4-mesh or the 7-mesh group individually (24 specimens each), every feature-set configuration achieves a *negative* mean R^2 (4-mesh: -0.81 to -3.56 ; 7-mesh: -0.30 to -4.65), even for the configurations whose raw MAE looks numerically similar to the pooled result (0.14–0.26 kN). A negative R^2 means the model performs worse than simply predicting the group mean for every specimen. Rank correlation is correspondingly weak (the best Spearman achieved within 4-mesh is only 0.10; within 7-mesh, 0.46). A low mean-absolute-error number within a small, low-variance subgroup is not, by itself, evidence of a working predictive relationship, and this dataset provides a direct illustration of that gap between absolute error and explained variance.

4.7.3 Is the pooled corrosion-load correlation genuine?

The clearest illustration of the same point uses correlation directly rather than a fitted model. Pooled across both mesh families, terminal visible corrosion and terminal ultimate load are correlated at Spearman $\rho = 0.462$ ($p = 0.0009$, $n = 48$) — a relationship that would ordinarily be read as a moderate, statistically significant positive association. Stratified by mesh family, this relationship does not merely weaken: within the 4-mesh group it is statistically indistinguishable from zero ($\rho = -0.046$, $p = 0.83$), and within the 7-mesh group it is negative and borderline significant in the *opposite* direction ($\rho = -0.401$, $p = 0.052$). Figure 4.6 shows all three panels together; the phase’s own saved output table labels the pooled figure explicitly as a "confounded" reference for this reason. This pattern is evidence of substantial confounding by mesh/campaign group structure — the pooled trend is strongly influenced by between-group differences (the lower-mesh, higher-load group also happens to have lower corrosion), rather than demonstrating a direct effect of corrosion on load-bearing capacity. It does not, on its own, prove that no corrosion-capacity relationship could exist under other conditions, nor does it fully decompose the pooled association into its causal components; it shows only that the pooled correlation in this dataset cannot be used as evidence for one.

4.7.4 Robustness under campaign holdout, and a residual-correction attempt

Under leave-one-campaign-out — the most demanding regime available, and, per Section 2.6, a test of extrapolation across the mesh and chloride design space rather than ordinary generalisation — every feature-set configuration collapses to strongly negative R^2 (metadata-only: MAE 0.465 kN, $R^2 = -4.64$). This is consistent with, rather than contradicting, the pooled-versus-stratified finding above: a model trained entirely within one mesh/campaign design space has no basis for extrapolating to a design space it has never seen, regardless of which features it was given. A further attempt to recover some of this performance by adding a second-stage, mesh-aware residual correction on top of the metadata-only baseline was tested and *worsened* MAE, RMSE, and Spearman relative to the baseline alone — a negative result the phase’s own saved comparison records explicitly rather than omitting.

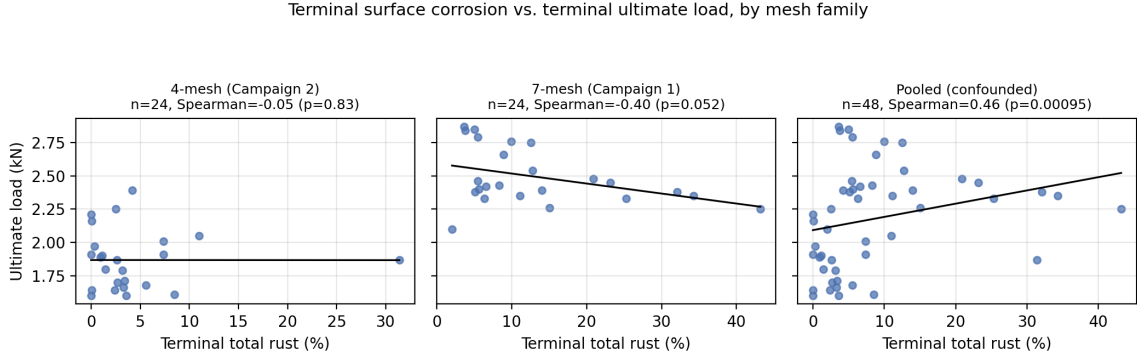


Figure 4.6: Terminal surface total rust percentage versus terminal ultimate load, stratified by steel-mesh family and pooled. The positive pooled association is not reproduced within either mesh family: it is approximately null in the 4-mesh group ($\rho \approx -0.05$) and negative in the 7-mesh group ($\rho \approx -0.40$). Generated directly from the phase’s own saved specimen-summary table.

4.7.5 Synthesis

Taken together, the evidence from this phase supports a specific and bounded conclusion: for terminal ultimate load, pooled across the observed specimen designs, specimen and exposure metadata is the strongest available predictor, and visible corrosion descriptors do not provide a reliable pooled improvement over it — with a narrow exception in a corrosion-onset subset that does not overturn the pooled result, and with within-mesh and cross-campaign evaluation both showing that whatever apparent relationship exists in the pooled data is confounded by, or fails to extrapolate across, specimen design. This is a materially more defensible scientific claim than either "corrosion predicts structural capacity" or "corrosion is irrelevant to structural capacity" — it is a claim about incremental predictive value, tested directly, under conditions designed to expose confounding rather than obscure it.

4.8 Four-Class Corrosion Classification and Augmentation

The final activity considered in this chapter is a later and structurally separate effort, addressing a different task from every phase above: direct image classification into an ordinal visible-corrosion severity category, using the raw-image deep-classification strategy introduced in Section 3.4 (strategy E). It targets the *current*, four-level severity schema established in Section 2.5 (Table 2.6) — not the five-level historical schema modelled in Section 4.4.1 — and is implemented as a standalone pipeline with no code dependency on any earlier phase.

The current schema is severely imbalanced: of 791 usable images, 658 (83.2%) fall in the lowest severity class, against 99, 20, and 14 images in the three more severe classes respectively (Table 2.2). This imbalance directly shaped the data-preparation work carried out so far. A five-recipe augmentation methodology — spanning brightness/contrast adjustment, saturation and colour-balance adjustment, blur and noise, a compound recipe, and a mirror-flip-plus-brightness recipe — was applied to every one of the 791 original images at a uniform six-fold expansion (one original plus five augmented copies), producing a *global offline augmentation inventory* of 4,746 images (791 originals plus 3,955 augmented copies; Table 4.5). Because the expansion factor is identical across classes, this inventory preserves the original class proportions exactly.

Specimens were then assigned to training, validation, and test partitions at the specimen level, consistent with the grouping principle of Section 3.2, with a leakage-safety rule enforced on top of that assignment: any augmented copy whose original image belongs to a validation- or test-assigned specimen is excluded from every split, so that validation and test contain only original,

unaugmented images (75 each) and no augmented copy of a held-out specimen ever reaches the training set. Because validation and test together account for 150 of the 791 original images, their 750 associated augmented copies (150×5) are removed from the global inventory before splitting. The resulting manifests therefore contain 3,846 training rows (641 original images plus 3,205 of their augmented copies), 75 validation rows, and 75 test rows — 3,996 rows used in total, not the full 4,746-image inventory. The training manifest’s own class distribution is 3,156/540/78/72 (82.1%/14.0%/2.0%/1.9%), close to but not identical with the global inventory’s proportions, since which original specimens — and therefore which of their augmented copies — fall into validation or test is not itself class-balanced. The majority class remains dominant throughout, and the project’s own documentation identifies class-weighting or an equivalent technique as a remaining requirement for whichever training procedure is used next.

Table 4.5: Global offline augmentation inventory (before leakage-safe exclusion of held-out specimens’ augmented copies), four-class severity schema. Every class is expanded by the same factor, so relative proportions within this inventory are unchanged; this table does not represent the final training-manifest composition (3,846 rows; see text). Source: the effort’s own dataset-preparation report.

Class (severity)	Original images	Augmented copies	Total	Factor
1 (lowest)	658	3,290	3,948	6×
2	99	495	594	6×
3	20	100	120	6×
4 (highest)	14	70	84	6×

A mirror (horizontal) flip is included among the accepted recipes here, even though the same transformation is explicitly excluded from the regression work in earlier phases (Section 3.4). The distinction is deliberate and physically grounded rather than inconsistent: total visible-corrosion severity, the target of this task, is position-invariant — a mirrored image has the same overall severity — whereas the strip-position and rust-location descriptors used in the regression phases encode left-right position on the specimen directly, and a flip would silently invalidate those features’ meaning. Several other augmentations were considered and rejected on physical-plausibility grounds: label-blending techniques such as MixUp and CutMix were excluded because the severity classes are discrete and cannot be meaningfully interpolated or partially overwritten; synthetic image generation was excluded as infeasible given fewer than twenty training images in the smallest classes; and elastic deformation, cutout, large rotation, vertical flip, and heavy colour shift were all excluded as producing image content inconsistent with how the specimens were actually photographed. Leakage safety is enforced by a runtime check in the splitting code, which raises an error if any augmented image is found in the validation or test partition, rather than relying on documentation alone.

The status of this effort should be stated plainly. Dataset preparation is complete: the target schema is defined, the class distribution is characterised, specimen-level splitting is implemented and verified, and the augmentation pipeline has been run to produce the expanded training pool described above. No classifier has been trained, compared, or evaluated on this data at the time of writing — no ResNet-50 or Vision Transformer fine-tuning, and no accuracy, macro-F1, or confusion-matrix result exists for this task in the repository. Data preparation for the four-class classification task is complete; classifier training and evaluation remain future work.

Chapter 5

Integrated Results and Comparative Analysis

Chapter 4 reconstructed the project chronologically, phase by phase. This chapter does not repeat that chronology. It instead integrates the evidence across phases to answer the scientific questions that recur throughout the project’s history: how reliably visible corrosion and hidden structural condition can each be estimated, what images contribute once metadata is already available, how conclusions change under stricter evaluation, what can and cannot be said about degradation and service life, and what the classification branch currently supports. Facts already established in Chapters 2–4 are cited by reference rather than re-derived.

5.1 Scope and Rules for Cross-Phase Comparison

The project’s successive phases differ in dataset basis (792 rows with one imputed observation versus 791 aligned rows, Section 2.5), target definition, feature representation, split construction, number of folds, model-selection procedure, use of manually assigned labels, and, for the categorical targets, the number of severity classes (Table 2.6). A lower MAE or higher R^2 in one phase than another does not, by itself, establish that one phase’s approach is superior: it may instead reflect a different, non-comparable protocol.

This chapter therefore distinguishes three kinds of comparison: a *direct numerical comparison*, permitted only where target, feature set, dataset basis, and evaluation regime are sufficiently aligned, with remaining protocol differences stated alongside the numbers; a *ratio comparison*, which normalises a result against its own phase’s baseline (e.g. a stricter-regime MAE divided by that same model’s grouped-holdout MAE) so only relative degradation, not absolute error, crosses phases; and a *qualitative comparison* of scientific conclusions, robustness patterns, and failure modes rather than metric values. Most comparisons below are of the second or third kind.

5.2 Visible-Corrosion Estimation

5.2.1 Current-corrosion estimation

Across the project, visible surface corrosion is the quantity estimated with the most consistent success. The classical baseline reached a grouped- holdout R^2 of 0.900 for current-week peak rust (Section 4.2); the interpretable multi-family feature phase reached $R^2 = 0.646$ (total rust) and $R^2 = 0.784$ (peak rust) under the same regime (Section 4.4.1). These results are not placed on a single ranked list, because they use different feature sets and, in the classical baseline’s case, a dataset basis and label-construction method that materially affects how the result should be read (next subsection). Taken together, however, they support a first, qualified conclusion:

image-derived representations of visible corrosion carry genuine, learnable signal under grouped-specimen evaluation.

5.2.2 Interpretability and the tautology risk

The strength of this signal must be read against two distinct label- reconstruction concerns, which implicate different targets and different degrees of severity and should not be merged into one claim. In the classical baseline, the current-corrosion (peak-rust) prediction is dominated (91.5% of feature importance) by a single rust-mask-derived colour-threshold feature closely aligned with the target (Section 4.2); this is a substantial shortcut/tautology risk, but the available evidence establishes only that this project-internal feature is closely aligned with the manually assigned label, not that it is computed by literally the same rule used to assign that label. In the robustness-quantification phase, by contrast, the near-identity is directly quantified: `surface_total_rust_pct` is correlated with a separate image feature at Spearman $\rho = 0.9996$ (Section 4.6) — close to near label reconstruction, a stronger claim than a shortcut risk. These two findings concern different targets (peak rust in the classical baseline; total/surface rust in the robustness-quantification phase), so they illustrate a recurring class of risk — label-adjacent features dominating a benchmark — rather than two data points about the same target. Peak rust in the later phases, for which no comparably near-tautological pairing has been found, remains the more meaningful test of genuine image-driven estimation, and this chapter treats these label-adjacent benchmarks as supportive rather than as headline evidence on their own.

5.2.3 Robustness

Robustness to distribution shift is where the picture becomes more qualified still, and where MAE, R^2 , and Spearman do not always tell the same story — and where the two phases that tested this do not agree with each other for treatment-level shift. In the interpretable-feature phase, the best-available regime model for each target remains positive under leave-one-campaign-out ($R^2 = 0.582$ and 0.595) but turns negative under leave-one-treatment-out ($R^2 = -0.677$ and -0.077 ; Table 4.2), evaluated over seven coarse treatment protocols (Section 4.4.2) — a genuine collapse in this phase’s own treatment-holdout evaluation.

The robustness-quantification phase’s own leave-one-treatment-out evaluation does *not* reproduce this fragility. Tracking each target’s own grouped-holdout-winning model under every regime — the appropriate same-model comparison, and how this phase’s own robustness table is itself constructed — its leave-one-treatment-out MAE is at or *below* its grouped-holdout MAE for both targets (ratios 0.55 and 0.76), mean R^2 is 0.995 (total rust) and 0.957 (peak rust), and Spearman correlation stays above 0.97 throughout, computed directly this session from the phase’s own saved per-fold outputs (Section 4.6). This phase’s leave-one-treatment-out evaluation also uses nine, more finely grouped treatment splits rather than the earlier phase’s seven coarse protocols, which may partly explain the difference; the evidence available here establishes the outcome, not the mechanism.

Leave-one-campaign-out robustness, by contrast, does replicate: both phases keep R^2 positive (or an MAE ratio at or below 1.0) for both visible-corrosion targets. Visible-corrosion estimation should therefore be described as robust to campaign-level shift in both phases; treatment-level robustness is a genuine failure mode in one phase but is not established as a project-wide limitation, since the other phase’s own treatment-holdout evaluation shows no comparable deterioration on any of the three metrics.

5.3 Structural-Condition Prediction

Under a grouped specimen holdout — the least demanding of the three regimes — hidden structural condition shows a modest but non-trivial relationship to the available features: wire-

area loss reaches $R^2 = 0.310$ and ultimate load reaches $R^2 = 0.385$ in the interpretable- feature phase (Section 4.4.2). This grouped-holdout result looks more encouraging than the strict-transfer results below for a specific, verified reason: it evaluates a random subset of specimens drawn from the same experimental mixture the model was trained on (Section 3.3.1), whereas the stricter regimes test extrapolation across specimen design or treatment protocol.

Under leave-one-campaign-out, structural performance collapses in both phases that tested it: mean R^2 falls to -0.39 (wire-area loss) and -8.36 (ultimate load) in the interpretable-feature phase, and the independently re-audited robustness benchmark shows ultimate load’s leave-one-campaign-out MAE inflating to 3.24 times its own grouped-holdout value, with Spearman falling to 0.35; wire-area loss’s leave-one-campaign-out Spearman turns negative (-0.089), worse than no rank information at all (Section 4.6). Within a single mesh family, the same collapse appears a third way, without invoking either stricter regime at all: every feature-set configuration tested for terminal ultimate load achieves a negative mean R^2 inside the 4-mesh or 7-mesh group considered alone, even where the raw MAE looks numerically reasonable (Section 4.7). Spearman correlation is what exposes this gap in every case: a model or a pooled correlation can report a plausible-looking absolute error while its rank correlation reveals that it carries little or no genuine discriminative signal once the sample is restricted to a design-homogeneous subgroup.

Two feature-importance diagnostics, computed independently in two phases, converge on the same explanation. The interpretable-feature phase’s full-data ultimate-load model is dominated by steel-mesh count and campaign indicators, not image descriptors (Section 4.4.2); the robustness-quantification phase’s five strongest univariate correlates of ultimate load are five metadata/temporal variables tied at exactly $|r| = 0.827$, a direct signature of their perfect alignment with campaign identity on the 48-row structural subset (Section 2.6, Section 4.6). This sparse-supervision constraint (48 rows, one per specimen; Section 2.4) is the most plausible reason structural inference is systematically weaker than visible-corrosion inference — a difference in available supervision, not primarily in modelling effort.

A further distinction matters for how the project’s own best evidence on this question should be read. The interpretable-feature phase’s structural models are mixed-input — metadata, manually assigned corrosion labels, and image descriptors together (Section 4.4.2) — whereas the robustness-quantification phase’s *final*, robustness-weighted structural model selection uses metadata alone for both targets: a RandomForest on metadata-only features for ultimate load, and a CatBoost on metadata-only features for wire-area loss (Section 4.6). The project’s own best-performing, most carefully selected structural models therefore use no image input at all. This is stronger and more specific than saying structural inference is weak: it is direct evidence about where whatever usable structural signal remains actually originates.

Bounded conclusion: the available evidence supports substantially weaker and less stable structural inference than visible-corrosion inference, driven by sparse supervision and campaign-linked confounding, and the project’s own final, robustness-weighted structural models do not use image input. This does not establish that structural prediction from corrosion images is impossible in principle — only that it is not established by the evidence gathered here.

5.4 Incremental Value of Image Information

A distinction the preceding two sections make necessary, and which this section addresses directly, is between two claims that are easily conflated: that images contain predictive signal, and that images add incremental predictive value once metadata is already available. The first is well supported for visible-corrosion estimation (Section 5.2). The second requires a design that isolates the image contribution from the metadata contribution — an image-only, metadata-only, and combined comparison under matched conditions (Section 3.5) — and this design was not applied consistently across the project’s history.

The deep-embedding phase compared only an image-only and a combined image-plus-metadata

configuration; no metadata-only baseline was trained (Section 4.3, Section 3.5). Its result — a lower validation error but a higher test error for the combined model relative to the image-only model — can therefore speak only to whether adding metadata to an image model helped in that one saved run; it cannot answer whether images add value beyond metadata, because the metadata-only anchor needed to answer that question was never computed. This chapter does not treat that phase as if it had run the complete ablation.

The terminal-load-focused phase is the only phase that made the full three-way comparison a designed, central part of its methodology (Section 4.7). Its pooled result, across all 48 specimens, is unambiguous: a metadata-only Ridge regression achieves the lowest MAE (0.173 kN) and the second-highest Spearman correlation (0.758) of every configuration tested, and adding RGB descriptors measurably worsens both MAE and R^2 (Table 4.4). A subset restricted to specimens with measurable terminal corrosion shows a narrower, real exception — a combined feature set edges ahead of metadata-only on MAE and R^2 there — but metadata-only still holds a marginally higher Spearman correlation even within that subset, and the phase’s own primary analysis is the pooled one, not the subset one (Section 4.7). This subset result is genuine evidence that images become more informative once corrosion has visibly progressed; it is not evidence that overturns the pooled conclusion.

Bounded conclusion: for terminal ultimate load — the only target with a complete three-way ablation — metadata is the strongest available pooled predictor, and visible corrosion descriptors do not provide a reliable pooled improvement over it, with a narrow, non-overturning exception once corrosion has visibly onset. That images carry some signal is not the same statement as that images add incremental value beyond metadata for this target, and the evidence supports the first claim more than the second.

5.5 Robustness and Distribution Shift

Section 3.3 defines the three evaluation regimes; this section asks what changed, across the project, when evaluation became harder — and where the two phases that tested more than one regime agree or disagree with each other. Leave-one-campaign-out is the regime that replicates most cleanly: structural targets are fragile under it in both phases, and visible-corrosion targets remain robust under it in both phases (Sections 5.2, 5.3). Leave-one-treatment-out is where the two phases diverge for visible-corrosion targets specifically: the interpretable-feature phase’s visible-corrosion models collapse under it, while the robustness- quantification phase’s do not (Section 5.2) — a phase-specific finding, not a replicated project-wide pattern. Structural targets, by contrast, are fragile under both stricter regimes in both phases, with leave-one-campaign-out typically the more damaging of the two — consistent with campaign identity, not treatment identity, carrying most of the design-linked confounding that structural models draw on (Section 2.6, Section 5.3). Because campaign holdout tests extrapolation across a design space in which mesh configuration, chloride level, and ageing duration all shift together (Section 2.6), while treatment holdout tests transfer within one design point to an unseen protocol, campaign-level fragility being the structural layer’s dominant weak point is a plausible consequence of *what* that regime perturbs.

Figure 5.1 shows both the replication and the divergence directly: the structural-target inflation pattern and the visible-corrosion targets’ campaign-level robustness recur independently across two phases with unrelated feature engineering, which makes it unlikely that these two patterns are artefacts of a single phase’s feature-engineering choices; the visible-corrosion treatment-level fragility does not recur, and should be read as specific to the interpretable-feature phase’s own treatment-holdout evaluation rather than as a property of the data itself.

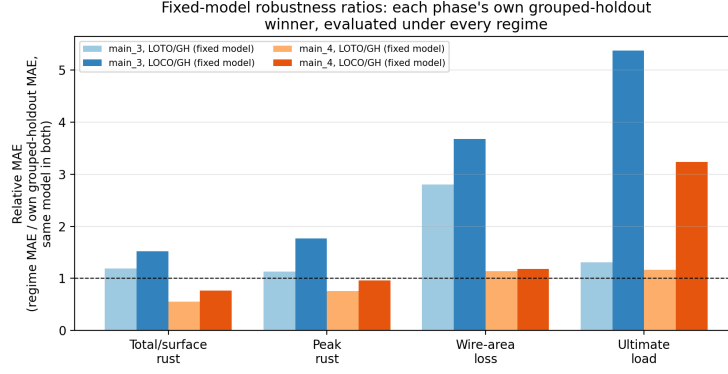


Figure 5.1: Relative MAE (regime MAE divided by that same model’s own grouped-holdout MAE) for the two visible-corrosion and two structural targets, using each phase’s own grouped-holdout-winning model evaluated under every regime — a fixed-model comparison, not a best-available-model-per-regime one.¹ No raw MAE from one phase is compared against another phase’s raw MAE. Structural targets inflate under both regimes in both phases, most severely for ultimate load under leave-one-campaign-out; visible-corrosion targets stay close to or below their own grouped-holdout MAE under leave-one-campaign-out in both phases, but diverge under leave-one-treatment-out — fragile in the interpretable-feature phase, robust in the robustness-quantification phase.

5.6 Degradation and Proxy-RUL Evidence

Two generations of the degradation/proxy-RUL derivation (Section 3.8) exist in the project, on configurations that are not directly comparable. The first (Section 4.5) fits a composite four-series derivation — two observed visible-corrosion series and two model-estimated structural series — against three threshold types over a 104-week projection horizon: of 48 specimens, 33 receive a finite estimate, 23 of which are exactly zero weeks, leaving only ten with a proxy-RUL that is both finite and genuinely forward-looking.

The second generation instead derives threshold-crossing status from a single series (model-estimated wire-area-loss fraction) against three absolute thresholds over a shorter, 365-day horizon — a narrower target and a materially different horizon, so the two are compared here only at the level of their conclusions, not their counts. Its own saved output classifies every one of the 144 specimen/threshold combinations tested as already past the threshold at the first observation, crossed during observation, or not crossed within the horizon: there is no genuinely forward-looking finite crossing anywhere in this refined output, a still more conservative result than the first generation’s ten forward-looking cases. The project’s own documentation separately flags, as an unresolved validation concern rather than a confirmed defect, that most specimens already show a model-estimated wire-area loss at or above the lowest threshold at their first observation — consistent with most crossings being attributed to the baseline rather than to projected degradation.

Bounded conclusion: both generations of the degradation/proxy-RUL pipeline demonstrate that the derivation chain defined in Section 3.8 can be executed end to end, and both generations’ own outputs argue against treating the resulting numbers as validated remaining-life estimates: the first because most of its finite estimates are degenerate, the second because it produces no forward-looking estimate at all under a more conservative configuration. True remaining useful life remains unavailable in this dataset regardless of which generation is consulted.

5.7 Current Four-Class Classification Branch

The four-class classification effort (Section 4.8) is structurally independent of every modelling question addressed above: it targets a different task (discrete severity classification rather than continuous estimation), a different, current four-level schema (Table 2.6), and shares no code with any earlier phase. Data preparation for this task is complete and verified: the target schema, the class-imbalance characterisation, the specimen-level grouped split, the leakage-safety runtime check, and the augmentation pipeline all exist and have been run. No classifier has been trained, compared, or evaluated on this data at the time of writing. This branch therefore contributes methodology and data-preparation evidence to this report, not model-performance evidence, and it is not compared against the interpretable-feature phase’s five-category classification results (Section 4.4.1), which model a different, historical schema with a different number of classes.

5.8 Integrated Findings

Table 5.1 summarises which quantitative results in this project may be compared directly, which may only be compared as ratios to their own baseline, and which must be compared only qualitatively. Table 5.2 then states, for each scientific question addressed in this chapter, the project’s current evidence level, main finding, main limitation, and the most defensible claim the evidence supports. Evidence-level terms are used as defined in the surrounding text of this chapter and are not numeric scores: *well supported* indicates a conclusion replicated across phases and stable under at least one stricter evaluation regime; *conditionally supported* indicates a conclusion that holds under some but not all tested conditions; *exploratory* indicates a working demonstration not yet validated against an independent standard; *not yet evaluated* indicates that no model-performance evidence exists at all.

Table 5.1: Cross-phase comparability. What may be compared directly, as a ratio to each result’s own baseline, or only qualitatively, and why.

Comparison	Permitted form	Reason
Interpretable-feature phase vs. robustness-quantification phase, same target family, robustness pattern	Ratio (regime MAE / own grouped-holdout MAE) or qualitative	Different feature sets and dataset bases (792 vs. 791 rows); absolute MAE not comparable, but both phases independently test the same robustness question.
Classical baseline vs. interpretable-feature phase, current-corrosion accuracy	Qualitative only	Different feature engineering, different label-construction tautology risk (Section 5.2); not placed on a shared ranked list.
Interpretable-feature phase vs. terminal-load-focused phase, ultimate-load MAE	Not compared numerically	Different feature sets, splits, and dataset bases (Section 4.4.2); comparing 0.121 kN against 0.174 kN as a leaderboard would be a category error.
Deep-embedding phase vs. terminal-load-focused phase, image value	Not compared numerically	Different targets (current-week peak rust vs. terminal ultimate load) and different representations; deep-embedding phase lacks a metadata-only baseline (Section 5.4).
First-generation vs. second-generation proxy-RUL	Qualitative (both support the same bounded conclusion)	Different target composition, threshold definitions, and projection horizons (104 weeks $\approx$ 728 days versus 365 days $\approx$ 52 weeks; Section 5.6).
Historical five-class classification (interpretable-feature phase) vs. current four-class classification	Not compared	Different category schema (Table 2.6); the current branch has no trained-model result to compare.

Table 5.2: Integrated scientific findings across the project. Evidence-level terms are defined in the surrounding text and are not numeric scores.

Question		Evidence level	Main finding	Defensible claim
Visible surface corrosion (Section 5.2)		Well supported (peak rust; campaign-level robustness); conditionally supported (total rust, tautology risk; treatment-level robustness, phase-dependent)	Robust under campaign holdout in both phases; treatment-level robustness diverges between phases (fragile in one, robust in the other)	Images carry genuine, learnable visible-corrosion signal, most credibly for peak rust, robust under campaign-level transfer; treatment-level transfer is unresolved, not established as a project-wide weakness
Hidden structural condition (Section 5.3)		Conditionally supported, weak	Grouped-holdout signal collapses under both stricter regimes; final selected models are metadata-only	The evidence supports substantially weaker structural inference than visible-corrosion inference; not established that images cannot help in principle
Terminal ultimate / incremental image value (Section 5.4)		Conditionally supported (against image value)	Metadata-only is the strongest pooled predictor; a narrow post-onset exception exists	Images do not add a reliable pooled improvement over metadata for terminal ultimate load
Cross-regime / cross-campaign robustness (Section 5.5)		Well supported (campaign-level pattern; structural fragility); conditionally supported (treatment-level visible-corrosion behaviour, phase-dependent)	Structural fragility under both stricter regimes, and visible-corrosion robustness under campaign holdout, replicate across two phases; treatment-level holdout behaviour for visible-corrosion diverges between phases	Robustness behaviour is target-family- and regime-specific; campaign-level and structural findings replicate across phases, but treatment-level visible-corrosion robustness does not and should be treated as phase-dependent
Degradation / proxy-RUL (Section 5.6)		Exploratory	Derivation chain executable end to end in two non-comparable configurations; both argue against forward-looking confidence	Useful as an exploratory screening framework; not a validated remaining-life estimate
Four-class classification (Section 5.7)		Not yet evaluated	Data preparation, splitting, and augmentation complete; no classifier trained	Contributes methodology evidence only; no performance claim can yet be made

Read together, these rows describe a project whose strongest evidence concerns visible-corrosion estimation, reinforced across independent phases except for treatment-level robustness, which is phase-specific rather than replicated (Section 5.5). Its most consequential negative findings were established in different ways and should not be flattened into one narrative of replication: campaign- confounded structural supervision was named early and reinforced by later, independent requantification (Sections 5.3 and 4.6); proxy-RUL’s non-validated status is reinforced by two independently configured generations of the same derivation (Section 5.6); but the absence of reliable incremental image value for terminal ultimate load rests on one phase’s own designed three-way ablation (Section 5.4), not on independent replication by a separate experiment, and is best read as a single well-designed test rather than a cross-phase-confirmed pattern.

Chapter 6

Scientific Interpretation and Research Decisions

Chapter 4 described what was built; Chapter 5 established what the integrated evidence shows. This chapter asks what those findings mean scientifically, and why they justified the major research decisions the project actually took. It does not repeat the chronological narrative or the metric-by-metric synthesis of the two preceding chapters; results are cited by reference, briefly, only where a specific number is the direct subject of an interpretive claim. This chapter distinguishes throughout between what the evidence supports observing and what may be inferred from it, and prefers hedged language ("is consistent with," "suggests," "cannot distinguish between") to causal claims wherever a causal mechanism is not established by the project's or the predecessor's own documentation.

6.1 From Visible Corrosion to Structural Condition: An Evidence Hierarchy

The project's successive research questions form an evidence hierarchy, not a sequence of independent tasks — and not a literal universal computational pipeline in which each stage is generated exclusively from the stage immediately above it. Figure 6.1 sets this out as a ladder of increasing inferential distance from directly observed quantities, decreasing supervision density, and increasing dependence on modelling assumptions; actual feature inputs at each stage are phase-dependent, not fixed by the ladder position alone. Direct observations and labels — images, visible-corrosion labels, and specimen/exposure metadata — sit at the top, densely available for nearly every observation (Section 2.2). Image representations and image-based estimates are computed from these observations. Hidden structural condition is a model's estimate, fit on 48 terminal measurements and applied to weeks at which no structural measurement exists (Section 2.4); its actual inputs vary by phase rather than depending on the image-representation tier alone — the interpretable-feature phase's structural models are mixed-input (metadata, manually assigned corrosion labels, and image descriptors together), while the robustness-quantification phase's final selections use metadata only (Section 5.3). Degradation trajectories are curves fitted to those estimated, not measured, series (Section 4.5). Proxy-RUL is a threshold-crossing time read off an extrapolation of one of those fitted curves (Section 3.8).

This hierarchy is offered as an organising explanation for a pattern already established in Chapter 5: results become progressively less robust and more heavily qualified moving down it. That pattern is consistent with supervision becoming sparser and model dependence compounding at each step; it is not, on its own, proof that any particular downstream quantity cannot in principle be estimated more reliably with different data or methods. The distinction matters for how the rest of this chapter should be read: each section below interprets why the evidence

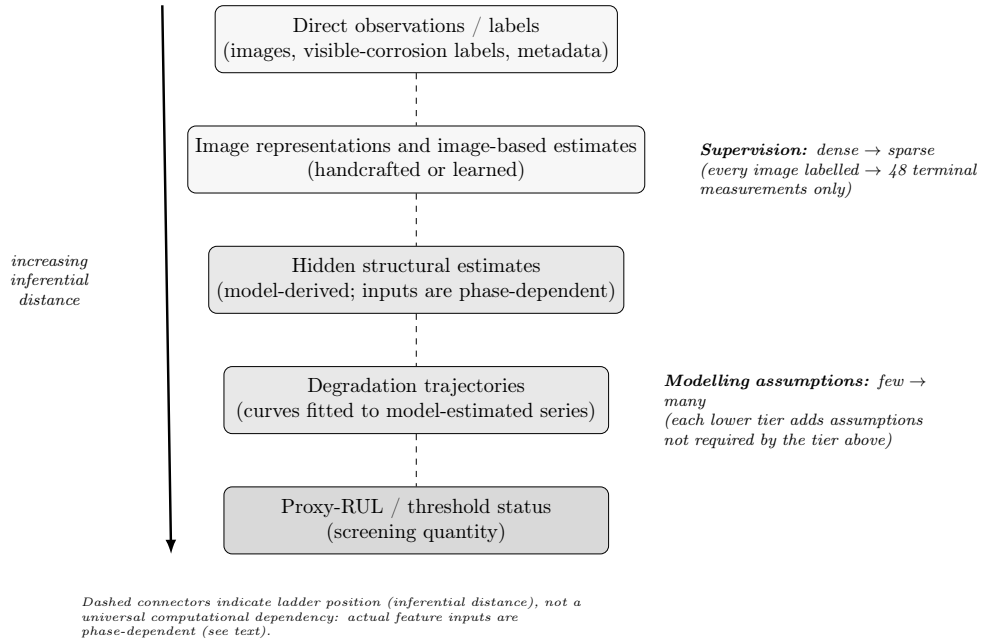


Figure 6.1: Conceptual report schematic (not a measured result). Position in the ladder reflects increasing inferential distance from directly observed quantities, decreasing supervision density, and increasing dependence on modelling assumptions — not a universal computational dependency in which each stage is generated solely from the stage immediately above it; actual feature inputs are phase-dependent, as discussed in the surrounding text. This illustrates why evidential confidence decreases downstream, not that downstream estimation is impossible.

looks the way it does, without treating the hierarchy itself as an argument for impossibility.

6.2 Why Visible Corrosion Was the Strongest Image-Based Task

Three independent phases, using unrelated representations (a fixed colour threshold, a frozen deep embedding, and an explicitly engineered multi-family feature set), each reached useful grouped-holdout accuracy for a visible-corrosion target (Section 5.2). This consistency across unrelated representations is consistent with the underlying task being genuinely well suited to image-based estimation: corrosion is a surface phenomenon that manifests directly in the photographic record, and it is the one target family labelled at the same fine time resolution as the images themselves (Section 2.2), giving every phase far more supervision to learn from than any structural target could offer.

This should not, however, be read as a uniform success story. Peak rust provides more meaningful independent evidence of genuine image-driven estimation than the total-rust benchmarks do, precisely because two separate phases independently found total/current rust predictions dominated by a feature closely aligned with, or nearly identical to, the label itself (Section 5.2). The lesson this supports is specific and worth stating plainly: high predictive accuracy is not sufficient evidence of a learned visual concept if the dominant feature substantially reconstructs the target’s own definition. This chapter treats that lesson as a general one, applicable beyond the two cases in which it was directly observed, not as a defect specific to those two phases. Robustness, similarly, does not transfer uniformly: campaign-level robustness for visible corrosion is a replicated finding, while treatment-level robustness is not (Section 5.5). The difference may partly reflect changes in treatment-group definition, size, or composition between the two implementations; the available evidence establishes the disagreement itself, not its mechanism, and does not rule out the visual task as a contributor.

6.3 Why Structural Inference Was Harder

Hidden structural condition differs from visible corrosion along every axis identified in Chapter 2: it is measured once per specimen rather than repeatedly, at a single terminal timepoint rather than throughout the observation period, for only 48 specimens rather than 791 observations (Section 2.4). This sparse, single-timepoint supervision is a major limitation consistent with the observed instability of structural models relative to visible-corrosion models, independent of any claim about the images themselves. It is compounded by the experimental design’s near-perfect alignment of campaign identity with steel-mesh configuration, chloride concentration, and ageing duration (Section 2.6): a model trained on all 48 rows has, in effect, only two campaign-level combinations of the principal design variables (mesh configuration, chloride level, and terminal duration) to learn from — treatment variation still exists within each campaign — and evaluating it under leave-one-campaign-out asks it to extrapolate to a combination of those variables it has never encountered (Section 3.3.3).

The available evidence supports a narrow, specific conclusion, and it is important not to overstate it in either direction. It would be wrong to conclude that surface corrosion does not affect structural capacity — no result here tests that physical claim directly, and Chapter 5 explicitly avoided asserting it. The evidence instead supports the narrower claim that this dataset and this experimental design do not isolate a robust, incremental, image-based relationship to hidden structural capacity. The distinction is between a physical relationship, which the available evidence can neither confirm nor rule out, and predictive identifiability from this specific dataset, which is what was actually tested and what the project’s results speak to. Reading the mixed-input character of the interpretable-feature phase’s structural models alongside the metadata-only character of the robustness- quantification phase’s final selections (Section 5.3) reinforces this reading: even when image descriptors are made available to a structural model, the model does not consistently rely on them once selection is made robustness-aware.

6.4 Metadata Dominance: Signal or Shortcut?

Metadata’s dominance of the terminal-load prediction task (Section 5.4) admits two readings that hold simultaneously rather than competing. Steel-mesh configuration, chloride exposure, ageing duration, and treatment protocol are genuine design and exposure variables with a plausible physical relationship to structural response; there is nothing suspect, in itself, about a model finding them predictive. At the same time, because these same variables are almost perfectly aligned with campaign identity in this sample (Section 2.6), a pooled model’s reliance on them cannot be distinguished, from the pooled evidence alone, from the model simply having learned which campaign a specimen came from. Both readings are consistent with the same pooled result, and the available evidence does not adjudicate between them from the pooled analysis alone.

This is why metadata should not be described as invalid or as a nuisance variable to be engineered away: it may well carry real information about structural response, and a metadata-only baseline is exactly the right comparison against which to judge whether images add anything further (Section 3.5). What the mesh-stratified and campaign- holdout evidence adds (Section 5.3) is a second, independent check on the same question, without relying on the pooled correlation at all. The failure to retain predictive quality within mesh groups and under campaign holdout weakens the evidence that the pooled metadata relationship is transferable beyond campaign-linked structure; it does not disprove a genuine physical metadata effect, only the claim that the pooled result demonstrates one. This is why pooled predictive performance can be genuinely useful — it is still the strongest available baseline for this task — while simultaneously being insufficient evidence, on its own, of a transferable structural relationship.

6.5 Why the Research Question Was Refocused

The shift from hidden-damage estimation toward the terminal ultimate-load refocus (Section 4.7) is best read as a scientifically motivated narrowing of claim scope, not as a retreat from a failed experiment. The hidden-damage chain — estimate structural condition at every observation week, fit a degradation curve to those estimates, extrapolate to a threshold crossing — compounds a separate layer of modelling uncertainty at each step (Section 5.6). Terminal ultimate load, by contrast, is directly measured for every one of the 48 specimens at a single, well-defined timepoint (Section 2.1), which removes the need for any intervening model-estimated quantity between the data and the target.

Reframing the question around that directly measured endpoint also made the resulting experiment more bounded and directly testable. "Do images add information beyond known design and exposure metadata for a directly measured terminal endpoint?" is a bounded, answerable question with a specific comparison built in from the start (Section 3.5). "Can a sparse hidden state be estimated at every intermediate week and then extrapolated into a remaining-life estimate?" is not incoherent, but is substantially harder to validate and diagnose, because disagreement at the final stage cannot be cleanly attributed to one component without independent intermediate ground truth. The refocused study's own design — matched feature-set comparisons, a mesh-stratified check, and a campaign-holdout stress test, all reported together rather than as a single isolated accuracy number (Section 4.7) — is consistent with this being a stronger experimental design, not merely a narrower one.

6.6 Scientific Value of the Negative Results

Several of the project's most useful findings are negative, and they should be read as evidence, not as failures to be minimised. That images do not provide a reliable pooled improvement over metadata for terminal ultimate load (Section 5.4) provides evidence against an otherwise plausible claim in this dataset, under the tested pooled setting, rather than merely failing to confirm it. That every feature-set configuration achieves negative R^2 within a single mesh family (Section 4.7) reveals a gap between absolute error and explained variance that a less careful evaluation would have missed entirely. That structural performance collapses under leave-one-campaign-out in every phase that tested it (Section 5.3) identifies exactly which evaluation regime future work in this dataset must budget for. That the refined proxy-RUL configuration yields a zero future-crossings result — no projected future threshold crossing among any of the tested cases, even though many are already crossed at baseline or during observation (Section 5.6) — is a more informative outcome than a handful of plausible-looking but unvalidated future estimates would have been. That the residual-correction attempt on top of the metadata-only baseline measurably worsened performance (Section 4.7) was recorded rather than discarded, which is itself part of what makes the surrounding positive claims credible. Each of these findings narrows the space of defensible claims, exposes a specific confound or data requirement, and motivates a specific next check — which is what a negative result is for.

6.7 What the Proxy-RUL Work Contributes

The proxy-RUL pipeline's contribution is methodological and diagnostic, not prognostic, and this chapter states that distinction directly rather than qualifying it away. Executing the four-step derivation end to end, twice, under two independently configured settings (Section 5.6), forced an explicit separation between observed and model-estimated condition series (Table 2.3) that a less disciplined pipeline could have left implicit. The refined generation's threshold-status taxonomy — crossed at baseline, crossed during observation, or not crossed within the horizon — is a more transparent way of reporting the same underlying uncertainty than a single ambiguous

"weeks remaining" figure would be, and its own zero future-crossings result is itself a diagnostic finding: the absence of any projected future crossing, together with the large number of already-baseline-crossed cases, is consistent with threshold status being strongly influenced by already-elevated baseline structural estimates, although that decomposition was not itself quantified. The physical plausibility of those baseline structural estimates remains, by the project's own account, an unresolved validation check rather than a settled fact. The project's own supporting documentation separately and explicitly recommends exporting genuinely held-out (out-of-fold) structural predictions in place of the current in-sample refits for any future degradation-modelling work — naming the relevant risk directly: that degradation and proxy outputs built on in-sample structural refits can appear more confident than the underlying structural model's own robustness evidence would justify. This is a downstream-optimism risk the project's own documentation makes visible, not one this chapter is introducing for the first time.

None of this amounts to validated remaining-life estimation, and the project's own dataset does not support it: true RUL validation would require either a recorded failure-event time or repeated, rather than single-terminal, structural measurements per specimen (Section 3.8), and neither is present. The pipeline's value lies in having built, tested, and critically examined the derivation chain that such data would require — clarifying exactly what is still missing — rather than in having produced a usable service-life number.

6.8 Interpretation of the Four-Class Classification Direction

The current four-class classification effort is a reasonable next direction on methodological grounds, independent of what a future trained classifier will show. It targets a directly observable quantity — image to severity category — rather than a model-derived or extrapolated one, using an explicit, current schema kept distinct from the historical five-level scheme used in the earlier dataset build (Section 2.5). Its class imbalance was characterised before any training was attempted (Section 4.8), its specimen-level grouped split and leakage- safety runtime check were implemented and verified rather than merely documented, and its augmentation recipes were chosen or rejected on stated physical-plausibility grounds rather than convenience. Read together, these choices show the branch incorporating, from the outset, the methodological lessons the earlier regression work had to establish one at a time over several phases (Section 6.9) — which is a reasonable basis for describing it as well prepared, whatever its eventual accuracy turns out to be.

This should not be read as more than it is. No classifier has been trained, compared, or evaluated on this data, and no ResNet-50 or Vision Transformer experiment exists yet in this project (Section 5.7). The evaluation practice already specified for whichever training procedure is used next — macro-averaged F1 and per-class recall in preference to overall accuracy, given the severity of the class imbalance, and a test partition left untouched until final evaluation (Section 3.7) — is stated here as the standard this branch should be held to, not as a result already obtained.

6.9 Methodological Lessons

Several lessons recur across the sections above and are worth drawing together briefly, as a closing statement rather than a scorecard. Evaluation design can matter more than model complexity: the project's regime comparisons changed which results were credible more than any change of model family did (Section 6.2, Section 3.3). A grouped holdout is necessary but not sufficient: it answers a narrower question than transfer to a new treatment or campaign, and the project's own later phases treated additional regimes as required, not optional (Section 3.9). Accuracy without attention to target provenance can be actively misleading, as the total-rust and peak-rust contrast shows directly (Section 6.2). A metadata-only baseline is not an optional control

but a precondition for any claim that images add value (Section 6.4). Sparse downstream labels bound what any model, however capable, can be shown to establish, independent of modelling effort (Section 6.3). Negative results are informative and should shape the next question rather than be minimised or omitted (Section 6.6). And where a directly measured endpoint exists, it is preferable to a longer chain of model-derived proxies for exactly the reason the terminal-load refocus illustrates: each additional derived step is an additional, separately uncertain modelling assumption (Section 6.5).

These lessons are offered as this project’s own accumulated methodological position, consistent with the evidence reviewed in Chapters 4 and 5, rather than as general claims about machine learning practice beyond this project’s scope.

Chapter 7

Software Engineering, Reproducibility, and Project Infrastructure

This chapter documents how the project’s codebase was organised and made progressively more traceable, and states plainly which concrete engineering issues currently limit a clean rerun of parts of it. It is not a tutorial on version control, configuration formats, or machine-learning tooling in general; every claim below is specific to this repository, with an exact file or artifact cited.

7.1 Repository and Pipeline Evolution

The project’s codebase falls into three engineering tiers. The exploratory prototype (`main_first`, Section 4.1) is a small set of unparameterised scripts with no configuration layer, no persisted metrics, and no held-out evaluation at all — its defining engineering weakness, not merely a scientific one. Four successive, increasingly structured modelling generations follow it: `main`, `main_2`, `main_3`, and `main_4`. The earliest of these four still parameterise thresholds and paths directly in training scripts. `main_3` introduces a single centralised configuration file (`configs/default.toml`) for its degradation and RUL thresholds. `main_4` completes this shift: six YAML configuration files (`dataset`, `features`, `modeling`, `thresholds`, `specimen_mapping`, `ultimate_load_refocus`) are loaded through one function and cached (`config.py`), so that a model-selection weight, a threshold, or a random seed is changed in one place rather than in scattered script arguments. `augmentation` is a later, separate branch built for the four-class classification effort (Section 4.8), following the same or greater configuration and leakage-safety discipline as `main_4`. Across all four modelling generations, source, configuration, and generated output become more consistently separated into distinct directories (`src/`, `configs/`, `outputs/`), and metrics, split manifests, and diagnostic tables are persisted as saved artifacts rather than console output.

7.2 Data and Artifact Provenance

The mature pipeline’s artifact chain runs from raw workbook and image files, through an aligned/canonical data table, to extracted features, saved split manifests, model metrics and predictions, and finally to the figures and tables cited in this report. This chain is traceable but not uniformly automated across every phase: `main_4`’s `outputs/` tree separates `data/`, `models/`, and `diagnostics/` consistently, while the earlier `main` and `main_2` phases save flatter `reports/` directories with less internal structure. The single canonical source for campaign, treatment, mesh, and chloride facts is `configs/specimen_mapping.yaml` (Section 2.1), used in preference to raw workbook columns precisely because it is a single, version-controlled table rather than

a value re-derived independently in each script. This report’s own `evidence_log.md` and per-chapter evidence-plan files extend that same provenance discipline one level further, recording the exact source file, tier, and caveat behind every number cited in Chapters 4–6.

7.3 Split and Evaluation Reproducibility

Whether another researcher can identify which observations belong to which partition — not what the partitions mean scientifically, already covered in Section 3.3 — depends on concrete, checkable artifacts. The classification/augmentation effort’s train, validation, and test manifests are saved as explicit CSV files (`Data/splits/{train,val,test}_manifest.csv`), and specimen assignment to validation and test is fixed by hardcoded, disjointness-checked specimen-ID lists rather than by a random draw; `make_splits.py` raises a `RuntimeError` if those lists overlap, if any augmented image is found outside the training partition, or if a specimen appears in more than one split. The regression phases’ grouped-holdout, LOTO, and LOCO folds are reproducible in a weaker sense: `main_4` records an explicit `random_state: 42` in its modelling configuration files, and per-fold split summaries are saved under `outputs/splits/`, but the fold assignments themselves depend on the installed scikit-learn version’s group-splitting implementation rather than being persisted as an independent manifest the way the classification splits are.

7.4 Image and Feature Reproducibility

The corrupted-image handling already documented in Table 2.5 is itself a reproducibility fact worth restating in engineering terms: three different policies (drop, impute, exclude) were applied across five implementation phases, so the row count a script produces depends on which phase’s code is run, and this is a version-specific property of each phase rather than something to harmonise after the fact. Feature provenance is otherwise traceable at the level of saved tables rather than individual descriptors: this report does not enumerate the ninety-one engineered features of the interpretable multi-family phase, and instead cites the saved feature tables and the automated feature-diagnostics report (Section 7.6) that characterise them in aggregate. Every new figure this report generates reads its input directly from a saved project CSV or parquet file, documented in `activity_report/figures/generated_scripts/` and cross-verified against the source table rather than transcribed by hand (Chapters 4–5, evidence log).

7.5 Configuration and Schema Issues

Two distinct, currently live issues were found in the repository’s source and configuration files, verified directly this session; a third, superficially similar pattern is a legitimate versioned change and must not be confused with either.

(A) A YAML boolean-parsing bug. `main_4/configs/specimen_mapping.yaml` records the no-treatment control protocol as an unquoted `treatment_coarse: NO`. Under PyYAML’s default resolver, the bare token `NO` is a YAML 1.1 boolean literal and is loaded as Python’s `False`, not the string `"NO"`. This is directly visible in the generated data: the `treatment_coarse` column of `outputs/data/master_table.csv` contains the literal string `"False"` for all 168 rows belonging to the ten no-treatment specimens, while the parallel `treatment_raw` column correctly reads `"NO"` for the same rows, and the `treatment_protocol` field (`no_treatment_control`) is unaffected throughout. This is a data-cleanliness and reproducibility defect in one derived column, not a scientific-result failure: every treatment-protocol table in this report (Table 2.1) was built from the unaffected fields, and the bug does not alter any already-reported count or result.

(B) A `"1"/"first" → "main_first"` string-substitution pattern, broader than previously logged. Wherever the current source tree should contain the standalone numeral 1 or

the word `first`, it frequently instead contains the string `main_first`. This was previously known only in two files, in category-range strings such as `"A_Total_Rust_Category_(1-4)"` appearing as `"(main_first-4)"`. Re-verification this session shows the pattern is substantially wider: it also replaces the pandas aggregation-function name `"first"` in named-aggregation and pivot-table calls across `main_2`, `main_3`, and `main_4` — directly confirmed to raise `AttributeError: 'SeriesGroupBy' object has no attribute 'main_first'` if such a call is executed today — and it replaces bare numeral hyperparameter values in `main_4`'s YAML configuration files (for example `n_jobs: -main_first, reg_lambda: main_first.0, glcm_distances: [main_first, 3]`), several of which would fail type validation if the affected pipeline stage were re-run unmodified. It also appears, harmlessly, in a package version string (`"0.main_first.0"`, which does not prevent the package from importing) and in comments, docstrings, and generated report prose. The pattern is confirmed present in `main`, `main_2`, `main_3`, and `main_4`, and confirmed absent from `augmentation`, the most recently written sub-project. Most affected files across `main_3` and `main_4` share an identical modification timestamp, and two files in `main` share a second, later timestamp. The saved scientific-output artifacts (metrics, predictions, and diagnostic tables) from these four affected pipeline generations that underpin the results reported in Chapters 4–6 were re-checked directly this session — more than two dozen files, dated 2026-03-06 to 2026-03-25 — and all predate both corruption timestamps. This does not extend to every configuration or documentation file in the repository: `specimen_mapping.yaml` itself carries the later corruption timestamp, though the specific fields this report cites from it (treatment protocol, mesh count, chloride level, campaign and terminal-week identifiers) are not among the affected tokens, which in this file are confined to a top-level version string and five narrative `series_description` fields not otherwise cited. This clustering is reported as an observation, not an explanation: the mechanism and cause of the substitution are not established by the available evidence and are not claimed here.

(C) The four-class/five-class category-schema difference (Table 2.6) is not an instance of (A) or (B). It is a verified, genuine change in the underlying workbook snapshot between two dataset builds, already established in Chapter 2, and should not be reinterpreted as a manifestation of the string-substitution pattern merely because both happen to involve small integers.

7.6 Model and Artifact Contract Limitations

`main_4/NEXT_STEPS_PLAN.md` independently lists five concrete engineering improvements; re-checking each against the current repository this session, rather than assuming the list is uniformly outstanding, gives a mixed picture. Saving the exact feature list used by each trained model is only partially done: one global list exists for the hidden-damage stage, but individual `best_model.json` files record only a target name and a model name, with no feature list or hyperparameters, and no equivalent file exists for the surface stage. Exporting genuinely held-out, out-of-fold structural predictions as the basis for degradation modelling is not done: the degradation stage still consumes in-sample hidden-damage refits (Section 5.6). An automatic feature-quality-control report, by contrast, is already implemented and run — `FEATURE_DIAGNOSTICS.md` is generated by `diagnostics.py` and covers exactly the checks recommended (near-constant, duplicate, and high-collinearity columns). Separating analysis tables from model-ready feature matrices is not done: the saved `surface_feature_table.csv` still mixes features, targets, and identifiers in one file. Saving raw alongside monotone-smoothed degradation trajectories, initially assumed outstanding, is in fact already done: `degradation_raw_vs_monotone_proxy.csv` stores both series together.

7.7 Reproducibility Improvements Already Achieved

Set against the issues above, several reproducibility practices are genuinely implemented, not merely recommended, and this chapter closes by stating them without overclaiming push-button reproducibility for the project as a whole. Implemented: the current 791-row alignment policy and its documented rationale (Section 2.5); the specimen-mapping table as a single canonical source; saved audit facts and split manifests for the classification branch; the automated feature-diagnostics report; a deterministic, hash-derived augmentation seed whose value is recorded in the augmentation pipeline’s own generated report; a leakage-safety runtime check enforced in code rather than documented as a convention; and this report’s own figure-generation scripts and evidence log. Recommended but not yet implemented: an exported dependency specification for `main_4`; correcting the two configuration issues in Section 7.5; and the two still-open artifact-contract items in Section 7.6. Given that live configuration and source corruption is confirmed present in `main_4` today, this chapter does not claim that the repository, as currently checked out, can be rerun end to end without modification — only that the specific gaps preventing this are identified, located, and, in most cases, narrow and fixable.

Chapter 8

Limitations

This chapter states the limitations of the evidence and implementation underlying Chapters 4–6, organised by category rather than by project phase. It does not re-run the results chapter: each item below is stated once, concisely, with a pointer to where it was established, and the consequence for what the report can and cannot claim.

8.1 Dataset and Experimental-Design Limitations

Every result in this report ultimately rests on 48 physical specimens tested across only two experimental campaigns, within which steel-mesh configuration, chloride concentration, and terminal ageing duration co-vary almost perfectly (Section 2.6). Structural supervision is available at a single terminal timepoint per specimen, with no repeated or intermediate structural measurement and no observed failure-event time (Section 2.4). The photographic record covers external surface appearance only; nothing in the dataset directly measures internal or subsurface condition at any non-terminal timepoint. Treatment groups are uneven in size, from two specimens in the smallest protocol to eighteen in the largest (Table 2.1). Together, these properties bound every downstream claim about structural condition, degradation, or transfer across campaigns or treatments to what a two-campaign, 48-specimen design can support — a limitation of the physical experiment this project inherited, not of the modelling choices made within it.

8.2 Label and Target Limitations

Visible-corrosion labels are manually assigned quantities (Section 2.2), not an independent instrumental ground truth, and two independent findings show that a benchmark built against them can be label-adjacent rather than purely image-driven: the classical baseline’s peak-rust prediction is dominated by a rust-mask feature closely aligned with the label (Section 4.2), and the robustness-quantification phase’s total-rust benchmark uses a feature correlated with its label at Spearman $\rho = 0.9996$ (Section 4.6). The category schema used for the two severity targets differs between dataset builds — five ordinal levels historically, four currently (Table 2.6) — a genuine version difference in the underlying workbook, not a data-quality error, but one that prevents any five-class and four-class classification result from being compared directly. Proxy-RUL thresholds (wire-loss percentages, a load ratio, a composite health-index cutoff, or the refined pipeline’s single-series thresholds) are engineering screening choices, not values derived from an independently validated failure criterion (Section 3.8), and no ground-truth remaining-useful-life value exists anywhere in the dataset against which any of them could be checked.

8.3 Generalisation Limitations

A grouped specimen holdout answers a narrower question than transfer to a new treatment or campaign (Section 3.3.1), and leave-one-campaign-out in this dataset is better understood as a severe extrapolation stress test than as an estimate of ordinary future performance, given how completely campaign identity is entangled with the principal design variables (Section 3.3.3). Treatment-level (leave-one-treatment-out) behaviour for visible-corrosion targets is not a settled project-wide property: it is a genuine failure mode in the interpretable-feature phase and is not reproduced in the robustness-quantification phase's own treatment-holdout evaluation (Section 5.2), and the report does not know why. Within-mesh subgroup analyses are based on only twenty-four specimens per group (Section 4.7), which limits how much confidence any single subgroup result — positive or negative — can carry on its own.

8.4 Structural-Modelling Limitations

Every structural claim in this report is built on 48 labelled specimens (Section 8.1); no amount of downstream modelling sophistication changes this. The interpretable-feature phase's structural models are mixed-input — metadata, manually assigned corrosion labels, and image descriptors together, not image-only (Section 4.4.2) — while the robustness-quantification phase's final, robustness-weighted selections use metadata alone (Section 5.3); no phase in this project has produced a robustness-weighted structural model shown to rely on image input. The degradation stage's structural estimates remain in-sample refits rather than out-of-fold predictions (Section 7.6), so no validated longitudinal structural state exists at any non-terminal week for any specimen — every such value in the dataset is model-derived, not measured (Table 2.3). As established in Chapter 6, this bounds what can be inferred to predictive identifiability from this specific dataset; it does not bound, and this report does not claim to bound, whether a physical relationship between surface corrosion and structural capacity exists.

8.5 Prognostic / Proxy-RUL Limitations

No true remaining-useful-life supervision exists in this dataset (Section 3.8), and this report's use of "proxy-RUL" throughout is a deliberate, consistently applied qualification of that fact. Every threshold used to derive a proxy-RUL or threshold-status value is a screening assumption, not a validated failure criterion (Section 8.2). Many specimens are already past the relevant threshold at their first observation in the refined pipeline's own accounting, and that same pipeline's own output shows *zero future* threshold crossings across every tested case — not zero crossings overall, since baseline- and during-observation-crossed cases are common (Section 5.6). No out-of-fold structural trajectory feeds either generation of the degradation model (Section 7.6), and no validation exists against an observed future failure or a repeated structural measurement of any specimen. None of this makes the pipeline useless: as Chapter 6 establishes, its value is methodological and diagnostic — it clarifies exactly what data a validated version would require — and that value is independent of, and unaffected by, its not being a validated prognostic method.

8.6 Current Classification-Branch Limitations

The four-class severity schema is severely imbalanced, with the majority class accounting for over four-fifths of images (Table 2.2). No classifier has been trained, compared, or evaluated on this data at the time of writing (Section 4.8), so no accuracy, macro-F1, or confusion-matrix result exists to limit. Uniform six-fold augmentation increases the absolute amount of training data without changing the relative class proportions, and does not by itself address the imbalance

(Section 4.8). Because no model has been trained, performance on held-out or external data is entirely unknown; this report does not speculate about the accuracy a ResNet-50 or Vision Transformer architecture might achieve, since neither has been implemented for this task.

8.7 Software and Reproducibility Limitations

Stated briefly here and detailed in Chapter 7, to which this section defers rather than repeats: a live YAML boolean-parsing bug and a broader "1"/"first" → "main_first" string-substitution pattern currently prevent an unmodified rerun of several `main_4` pipeline stages (Section 7.5); `main_4` has no exported dependency specification (Section 7.1); and two of the five artifact-contract improvements the project’s own documentation recommends remain unimplemented (Section 7.6). These are reproducibility limitations of the current source tree, not limitations of the scientific evidence itself: the saved scientific-output artifacts from the affected `main-main_4` pipeline generations that underpin the regression, structural, degradation, and robustness results reported here were generated before the earliest observed source-corruption timestamp (Section 7.5) and remain valid on that basis. The later `augmentation` branch, which underpins the four-class classification data preparation, is independently confirmed unaffected by this corruption (Section 7.5) and is not part of this claim.

Chapter 9

Current Status and Remaining Work

This chapter gives a clear snapshot of what is complete, what is prepared but not yet executed, and what future work the evidence in this report justifies. The three categories are kept distinct throughout: a workstream is *completed* only if it has produced the result it set out to produce; *prepared* if the groundwork is built and verified but the next step has not been run; and *recommended* if it is a justified next step that has not yet been scheduled or started.

9.1 Current Project Status

Table 9.1 summarises every workstream addressed in this report.

9.2 Immediate Next Work

Among the research and modelling workstreams — as distinct from completing this report itself, which remains the project’s immediate task (Table 9.1) — the highest-priority next technical step is training a baseline classifier for the four-class branch: it is the only such workstream that is fully prepared and has produced no model-performance evidence at all (Section 8.6). The practice this report recommends is already specified in Chapter 3 and is restated here as a checklist, not invented: keep validation and test specimen-disjoint, as already enforced by the saved split manifests (Section 7.3); select any hyperparameter or architecture choice using the validation partition only; evaluate the test partition once, at the end, and report macro-averaged F1 and per-class recall alongside a full confusion matrix rather than overall accuracy alone, given the severity of the class imbalance (Section 3.7); and apply class weighting or an equivalent technique, since the existing augmentation does not itself rebalance the classes (Section 8.6). A convolutional or transformer-based architecture (for example a ResNet-50 or a Vision Transformer) is a candidate choice consistent with the strategy already named in Chapter 3, not a committed or mandatory one; a simpler baseline should establish a reference point before any more complex architecture is attempted.

9.3 Structural / Prognostic Work Required for Stronger Claims

Model-engineering improvements and new data collection address different gaps and should not be conflated. On the engineering side, exporting genuinely out-of-fold structural predictions for the degradation stage, persisting a complete feature contract with every trained model, and checking the baseline structural estimates’ physical plausibility are concrete, currently unimplemented improvements (Section 7.6). The project has already run an explicit image-only/metadata-only/combined feature-set ablation for terminal ultimate load (Section 4.7); the corresponding ablation has not been repeated for the hidden-damage/degradation stage specifically, but this

Table 9.1: Current status by workstream. "Completed" means the workstream produced the result it set out to produce, including a negative or bounded one; it does not mean every question in that area is closed.

Workstream	Status	Evidence/output	Next action
Dataset audit/alignment	Completed	791-row aligned basis, corruption-handling audit (Table 2.5)	None required
Visible-corrosion regression	Completed	Three independent phases, campaign-level robustness confirmed (Section 5.2)	None required
Structural-condition feasibility	Completed (feasibility study)	Mixed-input and metadata-only models, both regime-tested (Section 5.3)	None required for this question as posed
Robustness diagnostics	Completed	Two independent phases, three regimes each (Section 5.5)	None required
Terminal ultimate-load refocus	Completed	Three-way ablation, subgroup and campaign checks (Section 4.7)	None required
Degradation / proxy-RUL screening	Completed (methodological/diagnostic)	Two independently configured generations (Section 5.6)	Further methodological refinement possible with existing data (Section 9.4); validated prognostic claims require additional longitudinal/failure data (Section 9.3)
Four-class classification	Prepared, not started	Schema, split, leakage checks, augmentation complete; zero trained models (Section 4.8)	Train a baseline classifier (Section 9.2)
Software / reproducibility cleanup	Mixed	Some items done (feature QC, raw/smoothed trajectories); YAML bug and string corruption still live (Chapter 7)	See Section 9.4
Final report/documentation	In progress	Chapters 2–8 drafted and frozen; Chapter 9 (this chapter) provisional pending this row's refresh; Chapters 1, 10, executive summary, abstract, appendices not yet written	Continue per approved sequencing

report does not identify a distinct, specific recommendation for it beyond the general out-of-fold and feature-contract improvements just stated. None of these engineering items, however, closes the evidence gap identified in Chapter 6: sparse, single-timepoint, campaign-confounded structural supervision (Section 8.1). Stronger structural or remaining-life claims would require some combination of data this project does not have — additional independent campaigns; design factors such as mesh configuration and chloride level decoupled across campaigns rather than perfectly aligned; more structural-labelled specimens; repeated or intermediate structural measurements per specimen, rather than a single terminal one; and an observed failure or service-life event time, if remaining-useful-life is ever to be a directly supervised objective. These are stated as requirements the current limitations point to, not as work already planned or scheduled.

9.4 Reproducibility Cleanup

In order of concreteness rather than importance: quote the YAML `treatment_coarse:` `NO` value (or otherwise force it to load as a string) to eliminate the boolean-parsing bug (Section 7.5); locate and repair the "main_first" string-substitution pattern, prioritising the instances confirmed to break execution (pandas aggregation calls, numeric YAML hyperparameters) over the cosmetic ones (comments, the package version string); export an explicit dependency specification for `main_4`, matching the practice already followed by `main`, `main_2`, `main_3`, and `augmentation`; persist a complete feature list and hyperparameter record with every trained model, not only the hidden-damage stage's existing partial list; and separate the remaining analysis tables (for example `surface_feature_table.csv`) from a model-ready feature matrix. None of these require new data collection; all are confined to the existing codebase.

9.5 Recommended Validation Before External Claims

Before any claim from this project is used to support a decision outside the project itself, this report recommends: validating any structural or robustness conclusion against at least one additional, independent campaign rather than the two available here; completing a strict, single-pass test evaluation for the classification branch once a classifier exists, with the test partition never consulted before that point; regenerating degradation and proxy-RUL outputs from genuinely out-of-fold structural predictions before interpreting those trajectories as genuinely out-of-sample estimates — a separate requirement from, and not a substitute for, the next one: explicitly justifying every proxy-RUL threshold against an independent domain or physical criterion, since out-of-fold structural inputs would improve the trajectories the thresholds are applied to without themselves validating the threshold values; repeating the subgroup and campaign-holdout robustness checks already applied to terminal ultimate load for any new target considered; and, where feasible, replicating key findings against an external or independently collected dataset. This report does not present the project, in its current state, as ready for deployment or for generalisation beyond the two campaigns studied.

Chapter 10

Conclusions and Next Steps

This chapter closes the report by returning explicitly to the research questions posed in Chapter 1, stating the activity’s overall scientific conclusion, and setting out what should be done next. It is a synthesis, not a further results chapter: no new evidence is introduced here, and every claim below is a restatement, at a bounded level of confidence, of a conclusion already established in Chapters 4–9.

10.1 Answers to the Research Questions

RQ1 (visible-corrosion estimation). Visible surface corrosion is the strongest and most consistently learnable image-based quantity in the available dataset, reached with useful accuracy by three distinct representations in turn (Chapter 4). This should not be read as unqualified: a total-rust benchmark in particular carries a label-adjacency risk that peak-rust benchmarks do not (Chapter 5), and campaign-level visible-corrosion robustness is supported across both later phases, whereas treatment-level behaviour is phase-dependent rather than uniformly robust or uniformly fragile (Chapter 6). The conclusion is bounded to this dataset and these representations, not a claim of universal generalisation.

RQ2 (hidden structural condition). Structural inference is substantially weaker and far less transferable than visible-corrosion estimation. The weaker structural results must be interpreted in light of the sparse, terminal, and campaign-confounded supervision available for those targets: structural labels are one measurement per specimen, while visible-corrosion labels are dense and repeated (Chapter 1, Chapter 2). One implementation’s structural models are mixed-input; a later implementation’s final, robustness-weighted structural selections use metadata alone; and campaign-holdout evaluation exposes severe transfer limitations in both (Chapter 5). None of this establishes that no physical relationship exists between surface corrosion and structural capacity — only that this dataset and design do not isolate one robustly (Chapter 6). The available experiments do not isolate the relative contribution of data limitations, image information, and model representation to this weakness.

RQ3 (incremental image value for terminal ultimate load). Under the pooled setting actually tested, images did not provide reliable incremental predictive value beyond specimen and exposure metadata (Chapter 4, Chapter 5). A post-onset analysis showed a limited improvement in some metrics once corrosion had visibly progressed, but it did not overturn the pooled result, and neither subgroup nor campaign-transfer evidence was strong enough to support a general claim that images add value. This conclusion is specific to the terminal ultimate-load target and the feature representations tested, not a general statement about image value elsewhere in the project.

RQ4 (evaluation robustness). Which evaluation regime is used materially changes what can be claimed. A grouped specimen holdout is useful for estimating generalisation to unseen specimens within the observed design mixture, but is insufficient on its own to support a transfer

claim. Leave-one-campaign-out is especially revealing for structural targets, but because campaign identity is entangled with mesh configuration, chloride level, and exposure duration in this dataset, it is better understood as an extrapolative stress test than as an ordinary estimate of future deployment performance (Chapter 2, Chapter 5). Treatment-level fragility in visible-corrosion estimation is a genuine finding in one implementation but is not reproduced in another, and should be treated as phase-specific rather than a settled, project-wide property (Chapter 5, Chapter 6).

RQ5 (degradation and proxy-RUL). The activity produced a working methodological and diagnostic threshold-screening pipeline, not a validated remaining-life prediction method. Its structural inputs are model-derived trajectories, not measurements; no true remaining-useful-life supervision exists anywhere in the dataset; and the refined pipeline configuration’s own output shows zero *future* threshold crossings — not zero crossings overall, since many cases are already crossed at baseline or during observation (Chapter 5, Chapter 6). Its structural inputs are not yet drawn from out-of-fold predictions (Chapter 7), which further limits how the resulting trajectories should be read.

10.2 Overall Scientific Conclusion

The activity began with a broad question: how much can be recovered about a ferrocement specimen’s condition from its repeated corrosion imagery alone? The evidence gathered across this report answers that question with a hierarchy of confidence rather than a single number. Visible surface condition is the quantity most reliably recovered from images. Hidden structural inference is markedly weaker, a result that is consistent with — and must be interpreted in light of — its sparse and campaign-confounded supervision, though the available evidence does not isolate how much of that weakness is attributable to the supervision itself as opposed to the image information or model representations used. Longitudinal degradation and proxy-RUL screening sit furthest from direct observation and carry correspondingly the least evidential weight, remaining exploratory and dependent on the model estimates beneath them (Chapter 6).

The central lesson this activity supports is not that one model outperformed another. It is that how strong a claim can legitimately be made is jointly determined by how directly the target is observable, how densely it is supervised, where the label itself came from, whether a metadata-only baseline was checked before crediting an image, and which evaluation regime was used to test the result. Read against that lesson, the project’s later decision to refocus the structural research question away from a sparse, multiply-derived hidden-damage estimate and toward terminal ultimate load — one of the two structural quantities measured directly, for every specimen, at a single defined timepoint, and the one selected for this analysis — was a scientifically sound narrowing of scope, not a retreat: it exchanged a harder-to-validate question for a more bounded, directly testable one, and answered it plainly.

10.3 Next Steps

Future work falls into three levels, kept explicitly distinct.

Immediate research step — prepared, not yet trained. The four-class visible-corrosion classification branch has complete, leakage-safe data preparation but no trained model. The recommended path is a simple baseline first, with any more complex convolutional or transformer architecture attempted only once that baseline exists; validation and test partitions must remain specimen-disjoint and the test partition untouched until a single final evaluation; and class-sensitive metrics, not overall accuracy, should be reported given the severity of the class imbalance (Chapter 9).

Methodological refinement with existing data — recommended, with key items still incomplete. Regenerating degradation inputs from genuinely out-of-fold structural predic-

tions, checking the physical plausibility of baseline structural estimates, completing the project’s model and feature-artifact contracts, and repairing the two live reproducibility issues identified in Chapter 7 are all achievable without new data. None of this would, on its own, make any remaining-useful-life claim validated; it would only make the existing exploratory pipeline more transparent and easier to audit.

Data required for stronger scientific claims — a future requirement, not scheduled work. The evidence gap this report has identified is ultimately a data gap, not a modelling gap. Materially stronger structural or prognostic claims would require additional independent campaigns, experimental design factors decoupled from campaign identity, more structural-labelled specimens, repeated structural measurements over time rather than a single terminal one, and — if remaining-useful-life is ever to be a directly supervised objective — observed lifetime or failure-event data. No degree of model refinement applied to the present dataset can substitute for this missing supervision.

10.4 Closing Statement

This activity does not deliver a deployment-ready predictor of structural condition or remaining service life, and it was not positioned to. What it does deliver is a progressively more rigorously evaluated account of what repeated corrosion imagery, in this specific experimental setting, can and cannot support — where the evidence is strong, where it is genuinely uncertain, and why — together with a progressively more reproducible and source-traceable basis on which the next modelling and experimental steps can be built.

Bibliography

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